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MILITARY HANDBOOK

NUCLEAR WEAPONS SYSTEMS, SAFETY DESIGN AND EVALUATION CRITERIA FOR

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DEPARTMENT OF THE NAVY
WASHINGTON, DC 20360

MIL-HDBK-272(OS)

NUCLEAR WEAPON SYSTEMS, SAFETY DESIGN AND EVALUATION CRITERIA FOR

1. This handbook, prepared by the Naval Sea Systems Command, was approved for printing and inclusion in the Military Standardization Handbook series.

2. This document provides valuable information and guidance to personnel concerned with the design and development of nuclear weapon systems and the evaluation criteria that is used to ensure nuclear safety.

3. Every effort has been made to reflect the latest information on nuclear weapon systems safety design and evaluation criteria. It is the intent to review this handbook periodically to ensure its completeness and currency. Beneficial comments (recommendations, additions, and/or deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Weapons Evaluation Facility (60), Kirtland Air Force Base, Albuquerque, NM 87117, with copies to Commander, Naval Sea Systems Command (SEA-643), Washington, DC 20362, and Commander, Naval Ordnance Station (Code 524), Indian Head, MD 20640, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Existing Department of Defense and Department of the Navy implementing instructions for nuclear weapon systems safety establish design requirements and evaluation criteria which are general in nature. These requirements must be analyzed to determine suitability and applicability for a given system development and procurement.

This document provides additional information and guidance to the project manager, acquisition engineer and others responsible for the design, test, and procurement of nuclear weapons components, equipments, subsystems, and systems on the nuclear safety design and evaluation criteria at various stages of their life cycle. The intent is not to provide detailed nuclear safety design and evaluation criteria, but rather to provide guidance and information which should be considered during design and procurement.

Every effort was made to assure that the referenced documents (paragraph 2) were in effect on the date of publishing this document. However, the currency of each document should be checked before use.

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ACRONYMS B-1

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1. GENERAL

1.1 Introduction

1.1.1 Scope. This handbook provides safety design and evaluation criteria which apply to nuclear weapon systems and their associated equipment for which funding and tasking are the responsibility of NAVSEA or appropriate PMs. This handbook should not be interpreted as a requirement to modify existing nuclear weapon systems which do not meet the safety criteria outlined herein. However, when an existing system is modified, the system should be evaluated to determine the extent to which these criteria should apply.

1.1.2 Purpose. The purpose of this handbook is to provide safety design and evaluation criteria for NAVSEA nuclear weapon systems and associated equipment. Applicable directives that pertain to the systems and equipment covered by this handbook are DOD Directive 3150.2, DOD Directive 5210.41, OPNAVINST C5510.83, OPNAVINST 8023.19, OPNAVINST 8110.18, OPNAVINST 8110.20, NAVMATINST 8020.2, NAVSEAINST 8020.10, NAVSEAINST 8110.3, NWP 14-1, and NAVFAC P-300.

This handbook contains extensive citations of reference documents such as MIL STD and MIL SPEC. In order to enhance the usefulness of this handbook, some attempt has been made to paraphrase certain essential and key information from cited reference documents. Additionally, in order to extend the useful lifetime of this handbook, new concepts and technologies which are likely to be used in new or modified nuclear weapon systems are considered. Examples include fiberoptics, integrated real-time processors, smart fuzing sensors, high energy lasers, reduction of human role, automatic target recognition and stealth technology. Because of the rapidly increasing requirements for software and firmware control associated with these new concepts and technologies, criteria dealing with software and firmware are addressed in considerable detail.

2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or requests for proposal, form a part of this handbook to the extent specified herein.

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SPECIFICATIONS

FEDERAL

KKK-T-2107	Truck & Truck Tractor, Commercial, Diesel or Gasoline Engine Driven 14,000 to 21,000 GVW, 4 X 2
KKK-T-2108	Truck & Truck Tractor, Commercial, Diesel and Gasoline Engine Driven 24,000 to 32,000 GVW, 4 X 2
KKK-T-2109	Truck & Truck Tractor, Commercial, Diesel Engine Driven 34,500 to 66,000 GVW, 6 X 4
KKK-T-2110	Truck & Truck Tractor, Commercial, Diesel or Gasoline Engine Driven, 16,000 to 35,000 GVW, 4 X 4

MILITARY

MIL-H-904	Hoists, Chain, Hand Operated, Hook and Trolley Suspension
MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems
MIL-M-8090	Mobility Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-H-19925	Hoists, Wire Rope, Electric Powered
MIL-T-21868	Trucks, Lift, Fork, Diesel, Shipboard and General Purpose General Specifications for
MIL-T-22292	Truck, Lift, Fork, Weapons Handling, Electric Powered
MIL-C-25200	Cable Assembly, Special Weapon, Electrical, General Requirements for

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MIL-T-25959	Tie Down, Cargo, Aircraft
MIL-T-27260	Tie Down, Cargo, Aircraft, CGU-1/B
MIL-C-28546	Cranes, Overhead Traveling, Underhung, Electric Powered
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-S-45152	Semitrailer, Lowbed, Commercial
MIL-T-45333	Trailers, Flatbed, 10 Ton 4 Wheel, M345
MIL-T-52932	Trucks, Lift, Fork, Internal Combustion Engine 4,000 - 6,000 pounds. Capacity, General Specification for
MIL-T-62032	Truck, Tank, Fuel Servicing & Aircraft Fuel Servicing, 1,000 to 24,000 Gal., Gas and Diesel Engine Driven, 16,000 to 44,000 pounds GVW, 4 X 2, 4 X 4, 6 X 4, & 6 X 6 Commercial
MIL-T-62318	Truck, Multistop Delivery, Forward Control Chassis, Gasoline and Diesel Engine Driven 13,000 to 21,000 GVW, 4 X 2
MIL-T-62319	Truck Cargo, Gasoline and Diesel Engine Driven, 14,000 to 24,000 pounds GVW, 4 X 2 and 4 X 4, Commercial
MIL-T-62323	Truck, Tank Sprinkler and Flusher Type 3000 Gallons Diesel Engine Driven, 34,500 to 43,000 pounds GVW, 6 X 4 and 6 X 6, Commercial

STANDARDS

FEDERAL

FED-STD-292	Truck, Light Commercial, Gasoline/Diesel Fueled 4 Wheel Drive 3,000 to 10,000 GVW

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FED-STD-307	Truck, Light Commercial, Gasoline/Diesel 2 Wheel Drive 3,000 to 11,000 GVW

MILITARY

Q-STD-328	Slinging and Tiedown Provisions for Lifting and Tieing Down Military Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristic, Measurement of
MIL-STD-463	Definitions and System of Units, Electromagnetic Interference Technology
MIL-STD-648	Design Criteria for Specialized Shipping Containers
MIL-STD-882	System Safety Program Requirements
MIL-STD-1365	General Design Criteria for Handling Equipment Associated with Weapons & Related Items
MIL-STD-1366	Material Transportation System Dimensional and Weight Constraints, Definition of
MIL-STD-1367	Packaging, Handling, Storage and Transportability Program Requirements
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields, General Requirements for
DOD-STD-1399	Interface Standard for Shipboard Systems Section 071 Part 1, D.C. Magnetic Field Environment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1512	Electroexplosive Subsystems, Electrically Initiated, Design Requirements and Test Methods

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MIL-STD-1574	System Safety Program for Space and Missile Systems
MIL-STD-1660	Design Criteria for Ammunition Unit Loads
DOD-STD-1679(Navy)	Software Development
DOD-STD-2105	Hazard Assessment Test for Navy Non-Nuclear Ordnance

HANDBOOKS

MIL-HDBK-5	Aerospace Vehicle Structures, Metallic Materials and Elements for
MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems
MIL-HDBK-255(AS)	Nuclear Weapons Systems, Safety, Design and Evaluation Criteria for

ORDNANCE DOCUMENTS

OD 44942	Weapon System Safety Guidelines Handbook

ORDNANCE PUBLICATIONS

OP 4	Ammunition Afloat
OP 3565 Vol. 2	Radio Frequency Hazards to Ordnance, Personnel and Fuel
OP 4098	Handling Ammunition and Hazardous Materials with Industrial Trucks

2.2 Other Publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SPECIAL WEAPONS ORDNANCE PUBLICATIONS

SWOP H-1	Navy Nuclear Weapons Handling Equipment, Certification Policies, and Maintenance Instructions with Illustrated Parts Breakdown

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SWOP 20-11	General Firefighting Guidance
SWOP 45-51	Transportation of Nuclear Weapons Material
SWOP 45-56	Instructions for Navy Movement of Nuclear Weapons
SWOP 50-20	Procedures for Preparation and Use of Stockpile-to-Target Sequence for Nuclear Weapons

NATIONAL SECURITY AGENCY PUBLICATIONS

NSAP KAG-30-A/TSEC (S-NOFORN)	Compromising Emanation Standard for Cryptographic Equipment

DIRECTIVES AND INSTRUCTIONS

DODDIR 3150.2	Safety Studies and Reviews of Nuclear Weapon Systems
DODDIR 5210.41	Security Criteria and Standards for Protecting Nuclear Weapons
OPNAVINST C5510.83	Navy Nuclear Weapons Security Manual
OPNAVINST 8023.19	Safety Criteria and Standards for the Movement of Nuclear Weapons by Non-Combatant Delivery Vehicles
OPNAVINST 8110.18	Navy Nuclear Weapons Safety Program
OPNAVINST 8110.20	Safety Studies and Reviews of Nuclear Weapon Systems
NAVMATINST 8020.2	Assignment of Responsibility and Authority for the Navy Nuclear Weapons Safety Program Within the Naval Material Command
NAVSEAINST 5100.13	Synthetic Fire-Resistant Hydraulic Fluids (MIL-H-19457 and MIL-H-22072), Precautions Relative to
NAVSEAINST 8020.6	Naval Explosives Safety Program; Responsibilities, Policies and Procedures for

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NAVSEAINST 8020.7	Hero, Policy for Conduct of a Safety Program to Alleviate
NAVSEAINST 8020.10	Nuclear Weapon Systems Safety Program Within the Naval Sea Systems Command
NAVSEAINST 8110.3	Nuclear Weapons Safety Program

MISCELLANEOUS PUBLICATIONS

NAVSEA S9086-1A-STM-000/CH-793	Naval Ships' Technical Manual (Security of Stowed Nuclear Weapons)
NWP 14-1	Loading and Underway Replenishment of Nuclear Weapons
NAVFAC P-300	Management of Transportation Equipment
NAVSEA S9AA0-AA-SPN-010/GEN SPEC	General Specifications for Ships in the United States Navy
AFSC Design HDBK DH 1-11	Air Transportability

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Definitions. The following definitions and acronyms are provided to clarify the usage intended for various terms in this document:

3.1.1 Abnormal environments. Abnormal environments are those environments as defined in the weapon's stockpile-to-target sequence document and military characteristics document in which the weapon is not expected to retain full operational reliability but must remain safe.

3.1.2 Access. Close physical proximity to a nuclear weapon or nuclear component in such a manner as to allow the opportunity to tamper with or damage a nuclear weapon when such tampering or damage could go undetected upon completion of the task being performed. Normally, a person would not be considered to have access if an escort or a guard were provided for either the person or the weapon when the person is in close proximity to the weapon.

3.1.3 Accident. Any unplanned occurrence involving loss or destruction of, or serious damage to, nuclear weapons or their components, which results in an actual or potential hazard to life or property.

3.1.4 Activate signals. Those signals developed and delivered to the switching assembly where they are combined with electrical power to produce control signals.

3.1.5 Armed. The configuration of a nuclear weapon in which a single signal will initiate the action required for obtaining a nuclear detonation.

3.1.6 Armed component. A weapon component is armed when it has responded to an influence to change its state to enable a nuclear detonation.

3.1.7 Armed weapon. A weapon is electrically armed when only a firing signal is required (all interrupters have been removed) to discharge electrical energy into the warhead detonators.

3.1.8 Arm/Disarm device. A mechanical or electro-mechanical device that provides a positive interruption of the firing circuit to prevent initiation of an explosive or pyrotechnic train prior to its commanded closure.

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3.1.9 Arming and Fuzing system. The aggregate of components within a nuclear weapon which generates signals to arm and fuze the warhead.

3.1.10 Arming system. That portion of a weapon which serves to ready (arm), safe, or resafe (disarm) the firing and fuzing systems and which may actuate devices in the nuclear system.

3.1.11 Automata. That class of sequential machines which, by alteration of internal state, are capable of performing logical, computational, or repetitive routines, i.e., automatic processors, computers, decoders, controllers, and their associated equipment.

3.1.12 Automation. Design and implementation of a system which is self-controlling.

3.1.13 Certified equipment. Items of support equipment, combat delivery vehicles, and noncombat delivery vehicles which have received a nuclear safety engineering evaluation and have been nuclear safety approved for use with nuclear weapons and identified in appropriate publications approved by an agency of the U.S. Navy.

3.1.14 Classified content. Information which makes memory a critical component.

3.1.15 Combat delivery vehicle. Any nuclear capable vehicle and its installed equipment used for the combat delivery of nuclear weapons. Operations of combat delivery vehicles are the subject of specified nuclear safety rules promulgated by the Chief of Naval Operations.

3.1.16 Command Disablement system. A system internal to the bomb/warhead or warhead container incorporating command and control features which destroys a weapon's ability to achieve a significant nuclear yield for a prescribed period of time.

3.1.17 Consent. A function implemented by a deliberate act that provides control over the select, prearm, launch, and fire functions.

3.1.18 Control current. Electrical current in a nuclear weapon used to change the functional status of a component associated with prearming, arming, or detonating the warhead.

3.1.19 Control signals. Those signals delivered to the nuclear weapon interface which result in prearming, safing, unlocking or release of a nuclear weapon.

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3.1.20 Coupling. A measure of data connectivity between a module and the program in which it is imbedded.

a. Common coupling. Modules are common coupled if they reference data held in a common pool or central repository for certain data structures accessible by all the modules.

b. Control coupling. Modules are control coupled if one module passes a flag or set of flags (control data) as arguments to the other, to directly influence the functioning of the receiving module.

c. Data coupling. Modules are data coupled if one module calls the other, if all input and output communications are in the form of arguments or parameters passed through the call-sequence interface, and if all such parameters are data (not control) elements.

3.1.21 Critical. The use in this handbook describes functions, circuits, activities, and hardware and software components which apply directly to, or control, the selection, loading, prearming, arming, firing, unlocking, releasing, launching, or targeting functions of a nuclear weapon system.

3.1.22 Critical component. A component which requires special handling during an identifiable portion or all of its life cycle. Examples of critical components are as follows:

a. A nuclear weapon, with or without its arming and fuzing system installed.

b. A combat delivery vehicle which has had all preload functions completed and is ready for nuclear weapon mate or load.

c. A combat delivery vehicle which has a nuclear weapon mated or loaded.

d. Hardware or software components so identified by the nuclear weapon safety study group.

3.1.23 Electrical isolation. Separation of electrical circuits, signals, or data to preclude ambiguity, interference, or information perversion. This may be achieved through physical isolation or by any property which distinguishes one electrical signal from all others (for example, time, phase, amplitude or frequency).

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3.1.24 Electrical interfaces. All electrical interfaces for the weapon system which are controlled by interface control drawings through joint Department of Defense and Department of Energy and Contractor subcommittees established by the Weapon System Project Officer.

3.1.25 Electroexplosive device. An electrically initiated explosive device having an explosive or pyrotechnic output.

3.1.26 Electromagnetic compatibility. The ability of electronic equipment, subsystems, and systems to operate together in their intended operational environments without suffering or causing unacceptable degradation because of unintentional electromagnetic radiation or response.

3.1.27 Electromagnetic interference. Any electromagnetic disturbance which interrupts, operates, obstructs, or otherwise degrades or limits the effective performance of electronics/electrical equipment. It can be induced intentionally as in some forms of electronic warfare, or unintentionally as a result of spurious emissions and responses, intermodulation products, and the like.

3.1.28 Electromagnetic pulse. The Electromagnetic Radiation from a nuclear explosion caused by Compton-recoil electrons and photoelectrons from photons scattered in the materials of the nuclear device or in a surrounding medium. The resulting electric and magnetic fields may couple with military systems to produce damaging current and voltage surges.

3.1.29 Electromagnetic radiation. Radiation made up of oscillating electric and magnetic fields and propagated at the speed of light. Electromagnetic radiation includes X-rays, gamma, ultraviolet, visible, and infrared radiation, and radar and radio waves.

3.1.30 Fail-safe. A design feature of a nuclear weapon system/component which ensures that, under failure, the item remains in the safe condition such that no critical functions, damage to equipment, or injury to personnel, will occur.

3.1.31 Firmware. Logic circuits in read only memory or specially protected portions of memory that may be altered by the software only under certain circumstances.

3.1.32 First article. The first production-configured item of equipment, which is generally used for evaluation testing.

3.1.33 Hard-wire. A dedicated discrete electrical circuit.

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3.1.34 Hazards of electromagnetic radiation to explosive ordnance. This is the danger of activating Electroexplosive Devices (EEDs) or otherwise activating ordnance due to radio-frequency electromagnetic fields of sufficient intensity.

3.1.35 Inadvertent. Due to human error, oversight, or equipment failure.

3.1.36 Incident. An unexpected event involving a nuclear weapon, facility, or component resulting in any of the following, but not constituting a nuclear weapon(s) accident:

a. An increase in the possibility of explosion or radioactive contamination.

b. Errors committed in the assembly, testing, loading, or transportation of equipment, or the malfunctioning of equipment and material which could lead to an unintentional operation of all or part of the weapon arming or firing sequence, or which could lead to a substantial change in yield, or increased dud probability.

c. Any act of God, unfavorable environment, or condition resulting in damage to the weapon, facility, or component.

3.1.37 Intent command signal. A control communication used to signify that the nuclear weapon is intended to prearm. The response of the weapon to this signal is to complete the prearming sequence, thus constituting a critical function.

3.1.38 Jettison. An authorized release of a weapon from the launch platform without initiation of critical functions.

3.1.39 Limited current. Monitor or test currents which are limited so that the most current that can be delivered to a nuclear weapon for monitoring or testing purposes will be less than that required to activate the most sensitive component in the arming sequence.

3.1.40 Limited life component. A component used in a nuclear weapon that decays with age and must be replaced on a periodic basis.

3.1.41 Logic circuits. Interconnected electrical, electronic, and microelectronic circuits that carry and control electrical impulses, for the purpose of providing the "ON-OFF" intelligence for predetermined relationships within a system or subsystem.

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3.1.42 Logistic transport. Those evolutions necessary to effect transfer of nuclear weapons or components from one nuclear capable activity to another within the Stockpile-to-Target Sequence (STS).

3.1.43 Masked weapon. The weapons projected operation trajectory that would interfere with any portion of the shipboard structure, antennas, adjacent launchers and similar equipment.

3.1.44 Military characteristics. A Department of Defense document submitted to the Department of Energy that specifies performance requirements and physical characteristics for a nuclear warhead, bomb, or basic assembly to be compatible with a specific weapon system or systems. This document is a contractural agreement.

3.1.45 Module. An identifiable, self-contained, compilable, independent set of computer instructions with one entry point and one exit point that fulfills some well defined function.

3.1.46 Monitor current. A limited current introduced into a nuclear weapon to determine the functional status of selected components.

3.1.47 Multiplexed system. A signal transmission system in which two or more signals share one transmission path or bus. A multiplexer is that portion of the multiplex system which formats transfers, and processes signals required for nuclear weapon monitoring, control, and separation.

3.1.48 Noncombat delivery vehicle. Any vehicle and its installed equipment used for the movement, loading, or transfer of nuclear weapons.

3.1.49 Normal environment. The expected logistical and operational environments, as defined in the weapon's stockpile-to-target sequence document and military characteristics document, in which the weapon is required to survive without degradation in operational reliability.

3.1.50 Nuclear cargo. A nuclear weapon or nuclear component (except limited life components) prepared for logistic movement.

3.1.51 Nuclear safety criteria. Design and evaluation criteria established as aids for ensuring that nuclear safety is a basic system engineering and procedural requirement in nuclear weapons systems and logistics systems.

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3.1.52 Nuclear safing. The prevention of a nuclear yield in the event of accidental detonation of the high explosive of a high explosive assembly weapon or ignition of the propellant of a gun assembly weapon.

3.1.53 Nuclear warhead. An item which normally consists of the explosive system, the nuclear system, and all electrical circuitry. It may be of an implosion type, gun type, or thermo-nuclear type. Nuclear components may be integrally contained. It is usually designed to be mounted in or on a projectile, torpedo, guided missile, rocket, bomb, or to its delivery system, or utilized as an atomic demolition munition.

3.1.54 Nuclear weapon. A device in which the explosion results from the energy released by reactions involving atomic nuclei, either fission or fusion, or both.

3.1.55 Nuclear weapon system. A nuclear weapon system consists of a nuclear weapon or nuclear warhead and those components required for its operation. A surface or subsurface weapon system normally consists of a missile or torpedo and associated launching and/or fire control system. Systems involving nuclear projectiles consist of the projectile and the gun to fire it. Systems involving nuclear demolition munition generally consist of the munition and the man or team and equipment associated with its placement and enablement on target. A nuclear weapon system may be supported by combatant and logistic ships, logistic aircraft, shore facilities, and by ancillary equipment as necessary to utilize the weapon.

3.1.56 One-point safe. A term used to describe the degree of safety in a nuclear weapon. A characteristic of a nuclear weapon which upon undergoing one-point detonation initiated anywhere in the high explosive system, has a probability of no greater than one in one million of producing a nuclear yield in excess of 4 pounds of trinitrotoluene equivalent.

3.1.57 Permissive action link. A device included in or attached to a nuclear weapon system utilized as a command and control device to preclude arming and/or launching until the insertion of a prescribed discrete code or combination. It may include equipment and cabling external to the weapon or weapon system to activate components within the warhead or weapon system.

3.1.58 Physical isolation. Physical separation of wiring, parts, modules, assemblies, etc., to assure no physical contact or interaction.

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3.1.59 Prearmed. The state of a weapon system in which a single launch, release, fire, or command signal will irreversibly initiate the arming and launch/fire/release sequence.

3.1.60 Prearming. A function(s) initiated or enabled by an operator(s) which places the weapon propulsion system in a pre-armed state.

3.1.61 Render safe procedures. The portion of the explosive ordnance disposal procedures involving the application of special methods and tools to provide for the interruption of functions or separation of essential components of unexploded explosive ordnance to prevent detonation.

3.1.62 Safe/Arm device. A device which provides electrical interruption of the firing circuits and mechanical interruption between the initiator and the subsequent explosive or pyrotechnic train.

3.1.63 Selection. The positioning of controls, switches and other devices, or a logic condition that places a nuclear weapon in a position or condition to be further prepared for launch, release, or emplacement. As used herein, "selection" is a critical function subject to nuclear safety criteria and evaluation.

3.1.64 Significant nuclear yield. The energy released through nuclear fission or fusion greater than the energy released by detonation of 4 pounds of trinitrotoluene equivalent.

3.1.65 Software. Information used to control or program, and which is processed or produced by automatic machines. It may be stored or transferred on a variety of informational storage media. Software, as used in the context of this instruction, includes firmware.

3.1.66 Stockpile-to-target sequence.

a. The order of events involved in removing a nuclear weapon from storage, and assembling, testing, transporting, and delivering it on the target.

b. A document which defines the logistical and employment concepts and related physical environments involved in the delivery of a nuclear weapon from the stockpile to the target.

c. This document is prepared in accordance with SWOP 50-20.

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3.1.67 Stop attack. The act of a single person to render a launched or fired weapon incapable of producing a nuclear yield (see 4.3.13).

3.1.68 Stray voltage. An unintended voltage existing in any part of a weapon system.

3.1.69 Structured design. A software hierarchial design technique with specified rules governing the hierarchy in terms of design order of modules, independence of modules, and specification of data and interfaces.

3.1.70 Structured program. A program whose control logic topology adheres to strict rules of form, being composed of interactions and nestings of a set of basic planar flowchart constructions: SEQUENCE, IF THEN ELSE, DO WHILE, DO UNTIL, CASE along with, perhaps, paranormal extensions.

3.1.71 Support equipment. The equipment ancillary to a nuclear weapon for the purpose of handling and maintaining the weapon as well as that associated test equipment required to verify the functioning of the nuclear system. (General and special purpose hand tools such as pliers, wrenches, screwdriver, and so forth, are specifically excluded).

3.1.72 Tactical movement. Those evolutions necessary to effect movement of nuclear weapons or components within the combat area; for example, within an amphibious objective area, and the area between communication zone and forward edge of battle area. Weapons will be in the configuration required by the tactical situation.

3.1.73. TEMPEST. Compromising emanations that are unintentional data-related or intelligence-bearing signals which, if intercepted and analyzed, can disclose the classified information transmitted, received, handled, or otherwise processed by electrical information processing equipment or systems.

3.1.74 Ultimate strength. That stress value at which a mechanical component undergoes structural failure.

3.1.75 Unauthorized launch. Any missile launch or projectile firing resulting from unauthorized acts. The missile/projectile need not follow a controlled trajectory, nor reach any specified impact point.

3.1.76 Volatile memory. A storage medium that loses information when power is removed from the system.

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3.1.77 War reserve (nuclear). Nuclear weapons material stockpiled in the custody of the Department of Energy or transferred to the custody of the Department of Defense and intended for employment in the event of war.

3.1.78 Yield point. That stress value at which a mechanical component first undergoes inelastic deformation.

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4. GENERAL REQUIREMENTS

4.1 Deviation Procedures. The design and evaluation criteria contained in this handbook cannot cover all possible situations. Innovative designs or advances in the state of the art may conflict with specific criteria in this handbook even though the design solution (or an alternative design) may seem to meet the objectives of the nuclear safety program or provide a significant safety advancement. In such cases, a deviation from the criteria may be requested from the NAVSEA appropriate PMs. A request for deviation must be supported by engineering data and complete proof of the alternative design (which includes an analysis of the safety provided). The request should be sent via the appropriate chain of command to:

To: Commander
Naval Sea Systems Command (SEA-643)
Washington, DC 20362

Copy to: Commanding Officer
Naval Weapons Evaluation Facility (Code 60)
Kirtland Air Force Base
Albuquerque, NM 87117

4.2 Nuclear Safety Standards and Certification

4.2.1 Nuclear weapons safety. Because of their political and military importance, their destructive power, their cost, and the consequences of an unauthorized or accidental nuclear or high explosive detonation, nuclear weapons will be protected against the risks and threats inherent in all their environments. The conservation of nuclear weapons as a national resource and the safety of the public, operating personnel, and property are of paramount importance. These weapons and weapon systems must be designed to incorporate maximum safety consistent with operational requirements. Within DOD, nuclear weapons safety is maximized by prescription and enforcement of basic safety standards. The implementation of these standards at the design level is accomplished by requirements for certain critical functions for weapons and/or systems. Nuclear weapons safety involving equipment which may be associated with nuclear weapons or warheads during their life cycle (such as storage and handling equipment) is implemented via requirements for nuclear safety certification of such equipment. Various DOD departments, commands, and activities which are charged with design, development, procurement, test, storage, handling, maintenance, or operation of nuclear weapons systems and/or components must ensure compliance with DOD safety standards by developing and

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enforcing nuclear safety design and evaluation criteria which apply to their respective systems, subsystems, or equipment.

4.2.2 Minimum standards. The criteria outlined here prescribe the minimum nuclear safety requirements for design evaluation of nuclear weapon systems and certification of associated equipment. These criteria are not considered to be design solutions or all-inclusive nuclear safety standards, and are not intended to restrict the designer in the methods and techniques used to meet operational design requirements. Because these requirements are not all-inclusive, the designer may still be obligated to introduce additional safety features as the specific situation dictates. These criteria do not invalidate other safety requirements of DOD publications. Moreover, Navy and contractor design activities covered by this handbook and by other requirements should be governed by whichever criteria is more stringent.

4.2.3 Nuclear safety standards. The DOD has established four safety standards contained in DOD Directive 3150.2 (which is implemented by OPNAVINST 8110.20) that are the basis for the nuclear safety rules governing nuclear weapon system operations and nuclear weapon system design and evaluation criteria. The system designer should bear in mind that these four nuclear safety standards must be met and that the operational use of a nuclear weapon system may be restricted if design safety criteria are not satisfied. These standards require that:

a. There shall be positive measures to prevent nuclear weapons involved in accidents or incidents, or jettisoned weapons, from producing a nuclear yield.

b. There shall be positive measures to prevent DELIBERATE prearming, arming, launching, firing, or releasing of nuclear weapons, except upon execution of emergency war orders or when directed by competent authority.

c. There shall be positive measures to prevent INADVERTENT prearming, arming, launching, firing, or releasing of nuclear weapons in all normal and credible abnormal environments.

d. There shall be positive measures to ensure adequate security of nuclear weapons, pursuant to DOD Directive 5210.41.

4.2.3.1 Controls. At the design level, these safety standards are implemented by control of critical functions in the sequence leading to detonation of the weapon. The safety features designed into NAVSEA nuclear weapon systems must provide positive control

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over the occurrence of these critical functions. As a minimum, the four critical functions described below must be included in the design of a nuclear weapon system.

a. Authorization or enabling. The first critical function to be controlled is authorization or enabling. The weapon system will use a device to control authorization. The function of the device will be to cause an enabled condition (a condition that permits selection, prearming, or arming) of the weapon system when authorization is received, through the command and control system, to prepare to use the system.

b. Intent command signal and prearming. The intent command signals the weapon that the personnel controlling the weapon want it to ultimately function and produce designed nuclear yield. The weapon's response to this command is to prearm. The prearmed condition is one of the inputs required by the weapon in its sequence of enabling, arming and firing. The weapon system design will keep the prearming function totally separate and distinct from the authorization or enabling function. Design features will preclude prearming in the absence of the intent command signal, and will also prevent any bypass of the prearming device(s) that would permit arming without prearming.

c. Launching or firing. The critical function of launching or firing will be controlled as follows:

1 Self-propelled weapons. Operation of the propulsion system (and control of launch) resulting in warhead movement will be controlled through two independent functions. The first function is a Safe/Arm (S/A) command for the ignition or motor start system. Without the command to arm, propulsion system ignition or motor start will not occur even if the ignition or motor start command is sent. In a rocket motor, a device with such a function is a rocket motor S/A device. The second function is an ignition or motor start command. Design features will preclude accidental transmission of either the arm command or the ignition/motor start command, and they will prevent any bypass of the ignition/motor start safing device so that ignition/motor start could not occur with the device safed.

2. Impulse launched weapons. Operation of the firing system will be controlled through two independent functions. The first function is an arm command for the firing system. Without the command to arm, initiation of the impulse will not occur even if the command to fire is sent. A device with such a function is called a firing system S/A device. The second function is a command to fire (firing command). Design features

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will preclude accidental transmission of either the arm command or the firing command, and they will prevent any bypass of the safing device, so that initiation of the impulse charge can not occur with the device safed.

d. Environmental sensing and final arming. Environmental sensing consists of the weapon comparing minimum environmental criteria (preset) with the environment it is actually experiencing. This function occurs after launch or firing of the weapon. Often, several environmental measurements are combined to improve the weapon's ability to discriminate against accidental or unauthorized arming and/or detonation. If the weapon is prearmed, the response of the weapon to sensing the required operational environment is final arming. Design features will include measurements of the environment so that non-operational environments are discriminated against to the maximum degree possible. If the weapon has self-contained guidance, a good guidance signal will be included as an element of the operational environment. The final armed condition is the condition that lets the selected fire signal (for example, radar, contact, or timer) detonate the warhead. Design features will preclude final arming unless the required operational environment is sensed, and they will prevent erroneous transmission of the required operational environment signal. Design features will preclude any bypass of the final arming device(s) that would permit nuclear detonation of the warhead without final arming.

4.2.4 Nuclear safety system support equipment certification. Nuclear safety certification policies contained in SWOP H-1 for handling equipment, vehicles, and procedures shall be followed.

4.3 General Design Criteria

4.3.1 Nuclear safety standards. The DOD nuclear weapon system safety standards (OPNAVINST 8110.20) must be met.

4.3.2 Single component malfunction. The malfunction or accidental operation of any single hardware or software component must not result in the selection, prearming, arming, launching, or firing of a nuclear weapon.

4.3.3 Human engineering. Designers must emphasize human engineering to minimize the probabilities of human error (see MIL-STD-1472). Using accepted human engineering methods, weapon system designers must conduct human error analyses and error reduction studies to identify any system mode(s) that may cause a hazardous condition. On the basis of these studies, designers

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must add features which will minimize human error and limit its consequences. The design must minimize the number of points within the system where human acts could degrade nuclear safety, and must stress positive measures to prevent any accidental operation of controls that could degrade nuclear safety.

4.3.4 Administrative procedures. Adequate nuclear safety must be designed into the system so that dependence on administrative procedures for safety is minimized.

4.3.5 Unauthorized access. The system must be designed such that a single individual with authorized or unauthorized access can not select prearm, arm, launch, or fire a nuclear weapon.

4.3.6 Protection of friendly territory. Weapon systems must be designed or controlled to prevent nuclear detonations except within specified target boundaries. Missile systems utilizing guidance signals must receive a good guidance signal from the guidance and control unit before final arming of the nuclear warhead can occur. The guidance signal must be withheld if a final guidance accuracy check indicates that the weapon will detonate outside specified target boundaries. As a design goal, the nuclear weapon system shall be designed to provide means or methods to preclude nuclear detonation in the event that a non-guided round will impact outside specified target boundaries.

4.3.7 Authorized access. Systems must be designed to permit authorized access (during all applicable phases of the Stockpile-to-Target Sequence (STS)) to those components and circuits as required to carry out command disable, emergency destruct procedures or to operate the permissive action link if applicable.

4.3.8 Environmental criteria. Nuclear safety design features must be considered over the full range of credible normal and abnormal environments and credible combinations thereof, to which all configurations of the system could be exposed. Specific normal and abnormal environmental parameters are system dependent and will be specified in the warhead or projectile STS document and in missile and carrier specifications.

4.3.9 Device reversibility. Propulsion system ignition/motor start and firing system safing devices, authorization devices, and prearm devices must be reversible in operation prior to launch initiation.

4.3.10 Administrative devices. Mechanical, electrical, or other types of devices that may be within a nuclear weapon system used for administrative functions (such as warhead

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identification, system status) shall be isolated from the various circuits and devices which control critical functions. The application of power (controlled energy or uncontrolled energy, e.g., lightning and induced currents) to an administrative circuit or device shall not cause a nuclear weapon to be selected, prearmed, armed, launched, fired, or jettisoned.

4.3.11 Safety requirement priorities. Safety requirements must be implemented according to the following priorities:

a. Safety requirements shall be assured through appropriate design features.

b. When safety requirements cannot be met by basic design, appropriate additional safety measures shall be used.

c. When the required degree of safety cannot be ensured through design or additional safety devices, special operational or administrative procedural restrictions shall be developed for the system or appropriate components.

4.3.12 Complexity of automated systems. The design of complex automated systems, intended to minimize or prevent human errors, shall not degrade nuclear safety.

4.3.13 Stop attack. If the design includes a "stop attack" feature, it will allow a single person to render a launched, or fired weapon incapable of producing a nuclear yield. The "stop attack" feature will be designed to be operable up to the moment of fuzing.

4.3.14 Explosive ordnance disposal. Nuclear weapons systems will provide for emergency access to those components and circuits required to effect render safe and disposal procedures.

4.3.15 Security requirements. Consistent with operational requirements, nuclear weapon systems security design must not inhibit nuclear safety features or requirements. Weapon system design must provide for security control to prevent or deter undetected access to nuclear weapon and nuclear weapon safety critical equipment or secure data. The design must ensure security control and protection of nuclear safety critical cards, tapes, keyboards, input controls, components, and devices which are developed to control or command the selection, loading, enabling, prearming, arming, firing, or launching of a nuclear weapon or nuclear weapon system.

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4.3.16 Hydraulic and pneumatic systems. Installed shipboard handling and storage equipment which utilizes hydraulic or pneumatic systems shall be designed in accordance with NAVSEA S9AAO-AA-SPN-010/GEN SPEC or applicable ship specification Sections 556 and 705. Positive control of weapons shall be exercised at all times during handling and storage, including both normal and casualty modes of operation. In the event of power loss, the equipment shall become immobilized until such time as safe, positively controlled operation can be resumed. Hydraulic fluids will be selected and utilized to minimize the hazards of fire and toxicity in accordance with NAVSEA S9AAO-AA-SPN-010/GEN SPEC or applicable ship specification Section 556 and NAVSEAINST 5100.13, respectively.

4.3.17 Nuclear biological chemical decontamination. Use of standard decontamination agents, such as chlorinated compounds, hydrocarbons, chelated compounds, and water-detergent solutions, shall not cause a degradation of safety through chemical reaction.

4.4 General Evaluation Criteria.

4.4.1 Responsibilities. The Navy organization or contractor responsible for each specific system or subsystem will develop the specific plans, tests, and analyses necessary to demonstrate the operation of nuclear safety features and the compliance with nuclear safety standards and requirements. Navy organizations that have responsibility for providing engineering and compatibility guidance for a system or subsystem developed by an allied country for use with U.S. nuclear weapons will be responsible for providing nuclear safety evaluation for such systems or subsystems. The responsible organization and the appropriate nuclear safety evaluation agency will determine the adequacy of test plans and completed tests and analyses as an integral part of the design and development process. The baseline for applying evaluation criteria is drawn from user requirements which are specified in formal documents such as the Justification for Major System New Starts, Operational Plan Data Document, Program Management Directive, the STS document, and the military characteristics document. Evaluation will be performed continuously by the responsible Navy agency as developmental modifications occur. General guidelines for evaluation requirements and responsibilities are provided by MIL-STD-882, and OD 44942. The DOD and Department of the Navy developed non-nuclear components which contain energetic materials, as defined in DOD-STD-2105, and are a part of the total nuclear weapon (exclusive of the DOE nuclear warhead) shall have a PMs sponsored systems safety program implemented as

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required by NAVSEAINST 8020.6. The details of such a program are described in OD 44942. The propulsion system shall be tested in accordance with DOD-STD-2105. This system safety program shall be reviewed periodically by the OPNAV Weapon System Explosives Safety Review Board prior to initial operational capability.

4.4.2 Use of existing specifications. When military specifications or standards which specify nuclear safety requirements exist, test and analysis requirements may be satisfied by demonstrating accomplishment of those specifications or standards. When operational requirements are found to deviate from the test requirements indicated in the military specifications or standards, the test requirements shall be supplemented accordingly.

4.4.3 Identification of safety features. Those safety features of subsystems and/or subassemblies of each item of equipment that provide nuclear safety shall be identified as a part of the evaluation process.

4.4.4 Control of targeting. Weapon systems will be evaluated to make sure their design is adequate to prevent accidental or unauthorized changes in targeting. They will also be evaluated to make sure adequate control is present to prevent nuclear detonations except within the boundaries of the designated target area.

4.4.5 Emergency access. Nuclear Weapon systems will be evaluated to make sure adequate emergency access is permitted to those components and circuits as required to carry out explosive ordnance disposal procedures. These systems must be analyzed to ensure that render safe and disposal procedures can be performed when necessary so as to positively preclude the possibility of a nuclear yield.

4.4.6 Arm/Disarm and Safe/Arm devices. Arm/Disarm and S/A devices shall be tested and evaluated in accordance with MIL-STD-1512.

4.4.7 Environmental criteria. The environmental requirements for each nuclear weapon systems are contained in the STS. The testing requirements for support equipment will generally be derived from the STS, the operational history of previous, similar equipment, and other equipment military standards..

4.4.8 Safety analyses, demonstrations, tests and reviews. Detailed criteria, design reviews, analyses, testing

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demonstrations, documentation, and reporting necessary to comply with the requirements of this document will ensure optimum incorporation of nuclear safety at the design level. Test, analysis, and evaluation require the application of sound engineering judgement by personnel responsible for nuclear safety in all phases of system development, acquisition, modification, and operation. The data derived from tests and analyses are essential to accomplishing nuclear safety evaluations. Both qualitative and quantitative analyses provide a basis for the nuclear safety evaluation.

4.4.8.1 Objective of safety analysis. The objective of the safety analysis is to identify potential weakness in the design of nuclear weapon procedures and nuclear weapon system critical components that may lead to an undesired event through design errors, storage and handling deficiencies, transportation hazards, or improper administrative or operational procedures. The safety analysis shall identify all the potential causes of nuclear safety hazards, examples of which are:

a. Accidental nuclear detonation.

b. An inadvertent or unauthorized event(s) such that any of the following may occur:

1. ED
2. Loss or theft
3. Selection
4. Loading
5. Prearming
6. Arming
7. Launching or firing
8. Sneak circuit
9. Jettison
10. Loss of power
11. CD

c. Any inadvertent or unauthorized event or sequence of events that may apply power to the nuclear weapon interface.

4.4.8.2 Examples of tests, reviews and analyses. The following types of tests, reviews, and analyses, as applicable, are examples of those required to support weapon system development of DOD equipment and safety study efforts, unless specifically exempted by NAVSEA or appropriate PMs:

a. Fault hazard analysis. Fault hazard analysis provides component level information on failure modes, effects, causes,

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and common modes susceptibilities within a given subsystem. The information is used in fault tree and common cause analyses.

b. Operating and support hazard analysis. The operating and support hazard analysis is used to identify and control maintenance and operations, personnel errors and procedures which can contribute to potential hazards. The resulting information is used in fault tree and common cause analysis.

c. Fault tree analysis. Fault tree analysis provides quantitative or qualitative measures of nuclear safety relative to hazards.

d. Common modes analysis. Common modes analysis, conducted after the fault tree is constructed, uses the fault tree to qualitatively access system susceptibility to potential failure.

e. Configuration definition. A review and list of all assemblies, wires, transmission lines, (including microwave and fiberoptics), connectors, and other hardware, which, when considered totally, contribute to potential hazards relative to the nuclear weapon and launching systems.

f. System interfaces. A safety analysis of the inter-relationships between the nuclear system and other systems.

g. Bent pin analysis. Bent pin analysis on connectors which transmit critical nuclear functions.

h. Electromagnetic radiation compatibility. Analysis of electromagnetic radiation compatability to include all applicable spectral regimes (e.g., optical if fiberoptics are employed in critical components or functions).

i. Failure mode and hazardous effects analyses. Failure mode and hazardous effects analyses of nuclear configuration components and subsystems whose operation or failure could cause critical nuclear functions to occur.

j. Preliminary hazard analysis. A preliminary hazard analysis will be conducted early in the design phase in order to perform a complete assessment of the hardware and software safety features. This analysis can provide the basis for determining other required analyses. See MIL-STD-1574 for areas to be considered.

k. System/subsystem hazard analysis. System/subsystem hazard analysis will be from the conceptual phase through post

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design. It will consist of general analyses of each subsystem and major component, their relationships to each other and their interfaces, environments which could affect them, and the hazards they could cause.

l. Non-nuclear system interface hazard analysis. Hazard analysis to identify potential or real hazards to the nuclear configuration or nuclear weapon as a result of a non-nuclear system operation or failure.

m. Logistic support analysis. Logistic support analysis to ensure that maintenance philosophy and procedures are safe.

n. Demonstration tests. Demonstrations (using production configured hardware) of the system's capability to meet design requirements, i.e., voltage, current, switching characteristics, and worst load conditions defined in the specifications.

o. Circuit isolation tests. Circuit isolation tests to make sure the nuclear configuration meets design criteria for electrical and electromagnetic isolation.

p. Electromagnetic interference and electromagnetic radiation tests. Tests will be conducted over all applicable spectral regimes to make sure onboard emitters and switching functions cannot initiate critical functions. For any ship or nuclear weapon (or both) which contains stealth components, testing must be accomplished with the entire configuration fully assembled (with dummy warhead) and installed.

q. Functional tests. Functional testing of the nuclear system on fleet ships. Induced failure tests will be considered to demonstrate the failure of critical components.

r. Software analysis. Provides assurance that the software or other program controls as designed, coded, and implemented, cannot contribute to accidental, unauthorized, or fault activation of critical nuclear weapon system functions.

4.4.9 Susceptibility to inadvertent or unauthorized actuation. The objective of inadvertent or unauthorized actuation analyses is to reveal facts about nuclear weapon systems design which will establish the basis upon which an overall determination can be made concerning the likelihood of such actuation. These facts may reveal design deficiencies in the system whereby either human actions, component failures, or combinations may negate the nuclear safety design safeguards. In reviewing these analyses, the Nuclear Weapon Safety Study Group must determine to what

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extent a nuclear weapon system is susceptible to inadvertent or unauthorized actuation.

a. Analysis of inadvertent actuation (equipment malfunction)

1. Assessment of potential events. Determine the malfunctions required for occurrence of significant undesired events using techniques such as fault tree analysis. The analyses shall determine the critical paths and identify equipment and components which could, by malfunction, result in the occurrence of such events.

2. Identification of safety features. Interlocks and other fail safe features to reduce or limit the likelihood of an undesired event caused by a malfunction shall be identified, described, and analyzed to determine level of effectiveness.

b. Analysis of inadvertent actuation (human error)

1. Critical areas of maintenance. Referring to those points of susceptibility previously identified, describe the general type, frequency, and degree of maintenance performed on the equipment. Designate those points as critical areas of maintenance.

2. Critical maintenance and operational procedures. Describe each maintenance or operational procedure that involves replacement, repair, or checkout of any component, module, drawer, or black box which can be identified as critical with respect to inadvertent actuation of the nuclear weapon system.

3. Identification of tools. Identify and describe the general types, quantity and complexity of tools and equipment used to perform maintenance on these components.

4. Fail-safe design features. Identify and describe each design feature or condition of the equipment which will cause the system to fail-safe if incorrect procedures are used by maintenance or operational personnel. Determine if the fail-safe feature(s) are adequate.

5. Human engineering design features. Identify and describe each design feature which, from a human engineering standpoint, is specifically designed to minimize human error.

6. Assessment of potential undesired events. From the information described above, arrange all maintenance activities in the order of their likelihood (highest to lowest) of causing an undesired event.

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c. Analysis of unauthorized actuation. An analysis of the weapon system to determine its susceptibility to unauthorized actions leading to events such as selection, loading, prearming, arming, launching or firing, jettison, command disablement, application of power to the nuclear weapon, and access, is required. This analysis shall, as a minimum, be based on the following threats:

1. One intruder agent who has authorized access to nuclear command and control communications or who would have the opportunity to gain access to nuclear command and control communications secure material. The intruder is assumed to have detailed system knowledge and unrestrained access to the operational weapon system.

2. Any number of intruders who are not authorized access to any part of the system, the classified information pertaining to the system, or communications associated with the system.

3. As many tools as can be carried by one person. It must be presumed, however, that once a person penetrates system security, a great number of tools could be at his disposal. Each specific threat must be treated uniquely, to establish a complete threat assumption.

4.4.9.1 Unauthorized actuation. The analysis shall identify those system components which can be tampered with or bypassed to contribute to or cause an unauthorized act and shall determine the following:

a. The preparations to be taken and procedures necessary to contribute to or cause an unauthorized use act (at each point identified to be susceptible). The time required for one person to complete the preparatory steps shall be estimated.

b. The tools and equipment necessary to accomplish the act.

c. The characteristics of associated shore, shipboard, and weapon system equipment, mechanical procedures, and other features which provide the ability to detect any attempt to penetrate the system.

d. The time required to detect such an unauthorized act and the time and methods required to thwart the attempt.

NOTE: From the information provided by the analysis, the methods of accomplishing an unauthorized

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event shall be arranged in order of most likely to least likely and the principal reasons for the order will be summarized. When documented, this information becomes highly sensitive and should be handled accordingly.

4.4.9.2 Susceptibility to nuclear yield generation due to accidents, incidents or jettisoning. Nuclear weapon systems shall be evaluated to ensure that the design precludes generation of a nuclear yield in all credible abnormal environments associated with accidents, incidents, or jettison events in which the weapon system may be involved.

4.5 Responsibilities

4.5.1 NAVSEA project office. The appropriate NAVSEA/PMs project office responsible for the development or modification of a specified nuclear weapon system shall utilize this document in the design and acquisition phase as an aid in ensuring nuclear safety.

4.5.2 Nuclear safety office. The NAVSEA nuclear safety office will assist and advise the NAVSEA/PMs project office in the implementation of nuclear safety analyses activities.

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5. DESIGN

5.1 Shipboard Systems

5.1.1 General. The design criteria set forth in this section apply to shipboard weapon systems as well as shipboard equipment and spaces used to stow, handle, transfer, assemble, load, and launch or fire nuclear weapons. In general, these weapon systems are (but not limited to) missile systems, gun fired systems, torpedo systems, and atomic demolition munitions. In addition, these design criteria apply to those shipboard systems that are used to monitor weapons and systems used for weapon control. Nuclear weapon systems that are ship-based but dependent upon an intermediate carriage mode by manned flight vehicle to the final launch point are to be designed in accordance with MIL-HDBK-255(AS).

5.1.2 Shipboard fire control systems. The design of shipboard fire control systems for nuclear weapon systems shall include positive means to prevent the unauthorized selection, prearming, arming, launching, jettisoning, or detonation of a nuclear weapon. Consideration shall be given to switches, devices, controls, and circuitry controlling the safe condition of a nuclear weapon as distinct and separate from those controlling conventional weapons. When possible, nuclear weapon designated switches and controls shall (1) be of a different design (shape and mode of operation) from any other immediately adjacent controls and switches, (2) require two or more distinct movements before they are activated, and (3) prevent any device, which singularly controls or selects nuclear weapons from being placed accidentally in the nuclear weapon position.

a. Other equipment. Conventional weapon system controls, switches, circuits, circuit breakers, and other devices, when operated in their intended mode, shall not cause the selection, prearming, arming, launching or firing of any nuclear weapon. The failure of any conventional weapon system control or device shall not permit selection, prearming, arming, launching, or firing of a nuclear weapon.

b. Fire control system power failure. The failure of power to the nuclear weapon system, fire control apparatus, switch, or circuit shall cause any prearmed nuclear weapon Safe/Arm (S/A) device (such as the unique signal switch and the weapon propulsion unit's S/A device) to return to the SAFE condition. In addition, any nuclear weapon select switch or control that is in the "select" position shall go to a "deselect" position. Restoration of power shall not automatically select, prearm, arm, launch or fire a nuclear weapon system.

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5.1.3 Launcher and launching devices. The launcher or launching device for a shipboard nuclear weapon shall be designed according to the following:

a. Single component failure. The failure of any single component in a launching system shall not cause the weapon to select, prearm, arm, launch, or fire the weapon.

b. Separate functions. The function within the launcher or launching device that controls the safe and arm system for the propulsion unit and the function that controls ejection or propulsion ignition shall be separate and distinct functions. The propulsion system shall not respond to an ignition or launch signal without having previously received a valid arm signal.

c. Accidental transmission. Launcher design features will preclude accidental transmission to the propulsion system of either the arm command or ignition command as a result of a component failure, stray voltage, electrical short, etc..

d. Launcher power failure. The loss of power to a launcher with a nuclear weapon present shall cause the USS and the propulsion system S/A device to return to SAFE. Return of power to the launcher shall not cause the S/A device to return to ARM.

e. Launcher motion. Motion of the launcher or launching device that is transmitted to an attached nuclear weapon shall not exceed the design limits of any single component, or group of components within the weapon. The motion induced into a nuclear weapon by a launcher shall not cause the propulsion system or weapon's warhead to prematurely prearm, arm, launch, fire, or release the weapon. When evaluating a new weapon for use on existing launchers, the weapon should be designed to existing launcher motions.

f. Physical limits. The total envelope of all possible positions that a launcher-weapon combination can occupy shall not permit physical interference between the weapon and shipboard structure.

g. Launcher-weapon separation trajectory. The launcher design shall include interlocks within the launch command (firing circuits) to prevent interference between the weapon's projected separation trajectory and shipboard structures. Consideration shall be given to ensure that such interference will be precluded for movable structures such as rotating antenna, adjacent launcher, and similar equipment.

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h. Masked weapon indicator. The design of the shipboard weapon system should include provisions for a visual indication at the fire control center when a weapon is obstructed by shipboard structure.

5.1.4 Assembly areas and magazines. Shipboard nuclear weapon assembly areas shall be designed to provide for the safety and security of nuclear weapons and nuclear weapon components. These spaces must be designed to permit at least two men to be present during authorized access, to prevent unauthorized entry, and to minimize the exposure of nuclear weapons and nuclear weapon components to abnormal environments. Shipboard nuclear weapon assembly areas and magazines shall be designed in accordance with SWOP 20-11, NAVSEA S9AA0-AA-SPN-101/GEN SPEC or applicable ship specification (Sections 703, 720, 730, 750, 770, 780, and 792), NAVSEA S9086-1A-STM-000 (Chapter 793), OP 4, 4098, and the following:

a. Sprinkler systems. Fire protection of the contents of magazines shall be designed to maximize coverage.

b. Personnel safety. Although magazines shall be designed to resist intrusion, these areas must provide means for personnel to exit the magazine rapidly in the event of an emergency.

5.1.5 Administrative devices. Mechanical, electronic, or other types of devices that may be within a nuclear weapon or within a nuclear weapon tactical container that are for administrative functions (such as PAL, warhead identification, indication of system status) shall be separate and distinct from the various circuits and devices which control critical functions. The application of power (controlled energy or uncontrolled energy, e.g., lightning strikes) to an administrative circuit or device shall not cause a nuclear weapon system to select, prearm, arm, launch, or fire.

5.2 Ground Delivered Systems

5.2.1 General. The design criteria described in this section applies to ground delivered nuclear weapon systems (artillery fired atomic projectiles, and atomic demolition munitions as examples).

5.2.2 Command control systems. The design of command and control systems for ground delivery of nuclear weapons shall include positive means to prevent the unauthorized launching, prearming, arming, firing, or detonation of a nuclear weapon.

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5.2.3 Delivery devices. Howitzers, cannons, launchers, and other devices used to launch or fire nuclear weapons shall be designed according to the following:

a. Single component failure. The failure of any single component shall not cause the weapon to prearm, arm, launch, or fire the weapon.

b. Separate functions. Those functions and devices within a delivery device that control the S/A system for the firing mechanism and those functions within the device that control the firing system shall be separate and distinct functions. The firing system shall not respond to a command to fire without having previously received a valid arm signal, or a deliberate manual operation.

c. Accidental transmission. Delivery device design features will preclude accidental transmission of either the arm command or the fire command as a result of a component failure, stray voltage, electrical short, etc.

d. Delivery device power failure. If applicable, the loss of power to a nuclear weapon delivery device shall cause the firing system S/A device to return to SAFE. Return of power to the delivery device shall not cause the S/A device to ARM.

5.2.4 Nuclear projectiles. Nuclear projectiles and associated equipment (hoist, loading trays, rams, etc.) will be designed so that the motion and forces induced into a nuclear weapon by this equipment does not cause the weapon to prearm, arm, or detonate prematurely.

5.2.5 Atomic demolition munitions. Delivery techniques and handling equipment (if applicable) will be designed so that the environmental conditions (motion and forces due to parachuting, ground impact, hydrostatic and barometric pressure variations, and other effects) will not cause the munitions to prearm, arm, or detonate prematurely.

5.3 Transportation (Logistic and Tactical) and Related Support Equipment

5.3.1 General. Safety will be incorporated into the design of vehicles and support equipment used to transport, store, support, load and unload nuclear weapons, where applicable, with respect to operational concepts. The vehicles and equipment will meet appropriate structural, environmental, and mobility requirements as defined in KKK-T-2107, KKK-T-2108, KKK-T-2109,

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KKK-T-2110, FED-STD-292, FED-STD-307, MIL-H-904, MIL-S-8512, MIL-H-19925, MIL-T-21868, MIL-T-21870, MIL-M-8090, MIL-T-22292, MIL-H-26935, MIL-C-28546, MIL-S-45152, MIL-T-45333, MIL-T-62032, MIL-T-62318, MIL-T-62319, MIL-T-62323, MIL-T-62333, MIL-T-52740, MIL-T-52932, MIL-STD-1365, MIL-STD-1366, MIL-STD-1367, SWOPs 45-51 and 45-56, and AFSC Design HDBK DH 1-11. The Stockpile-to-Target Sequence (STS) document will define modes of transportation. The intent of the criteria in this section is to prevent damage to the nuclear weapon during handling and transportation. The safety design factors will allow for the uncertainties in predicting operational conditions such as overloads, fatigue, wear, corrosion, residual stress, temperature influence on metal properties, and impact loads. The safety design factors will also allow for the uncertainties or variations in material strength and manufacturing techniques and the uncertainties introduced by simplified design and test procedures. These criteria supplement good industrial design practices, standards, and features. They are not intended to prohibit the use of any commercial design of equipment (such as trucks, truck tractors, semitrailers, trailers, and cranes) which meets these criteria.

5.3.2 Ground transportation and support equipment design criteria. The following criteria apply to ground transportation equipment used to transport nuclear weapons on their basic structure, including trailers, semitrailers, self-propelled ground vehicles, forklifts and weapon loaders:

a. Weapons support. This equipment will be designed so that the nuclear weapon is supported by the basic frame of the equipment rather than by lift arms, cables, or hydraulic systems. This requirement does not apply to equipment used only to position or transfer nuclear weapons in a designated area such as a weapons storage area.

b. Grounding provisions. Static grounding provisions will be provided for equipment designed for specific nuclear systems in order to prevent static electricity discharge through the weapons, if the design agency requires it.

c. Fire retardant/containment. The equipment will be designed to reduce the propagation of fire due to electrical or fuel system failure from transmission of mechanical shock to the nuclear weapon(s).

d. Tiedown configurations. The tiedown configuration for ground transport of all nuclear weapons must have and consider the following:

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1. Tiedown points. Provide the number of cargo tie-down points sufficient to restrain design loads.

2. Tiedown terminals. Ensure that the end of each tiedown device terminates at a separate attachment point without passing through any other attachment point.

3. Road conditions. Consider worst case road conditions as identified in the STS.

5.3.3 Structural design criteria. The following structural design criteria will be determined by:

a. Rated load. The rated load is that combination of forces that the basic equipment must support or resist in a static state. This static load consists of the weapon(s) and the associated handling and tiedown equipment.

b. Dynamic load. To determine the dynamic load, the designer must consider lateral, horizontal, and vertical dynamic loads, accelerations encountered in ground transport environments, and shock loads associated with mate, demate, load, and unload operations.

c. Design load. The design load will, where practical, be based on the rated load multiplied by a factor of three or on the dynamic load multiplied by a factor of two, whichever is greater. This design load will be considered the minimum load for attaining the design stress levels.

d. Allowable stresses. In determining allowable stresses for equipment, the designer will select the material and type of stress specified in government publications (such as MIL-HDBK-5) and national standards (such as those produced by the Society of Automotive Engineers or the American Society for Testing and Materials). In cases where both an average and a minimum stress are specified, the minimum stress will be used.

5.3.4 Design criteria for trailers and semitrailers. All trailers and semitrailers used to transport nuclear weapons will meet the specific criteria in section 5.3.1 and the following:

a. Free-wheeling support equipment. All free-wheeling support equipment will be designed to restrain itself, e.g., dead-man-brake, with its rated load.

b. Trailer allowable dynamics. In a fully loaded config-uration, trailer and semitrailer design will be designed to minimize:

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1. Tendency to yaw, sway, or skid under operating and braking conditions and minimum stopping distances.

2. Tendency to tip, tilt, or jackknife.

c. Parking brakes. Parking brakes will be designed to hold a fully loaded trailer on an 11.5 degree incline with the trailer headed up or down.

d. Other braking requirements. Trailers and semitrailers will be equipped with an emergency braking feature to bring the trailer or semitrailer to a controlled stop if breakaway from the tow vehicle occurs. The design will allow for a manual override of the breakaway device when the trailer is being moved by hand.

e. Mobility. Mobility requirements of MIL-M-8090 will be met.

5.3.5 Self-propelled vehicles. This subsection covers self-propelled ground vehicles (such as trucks, vans, tugs, tractors, and high-lift trucks) used to transport nuclear weapons and vehicles used to tow (as a prime mover) other vehicles carrying the nuclear weapons (excluding self-propelled howitzer). This subsection does not include loading/unloading trucks. Section 5.3.8 applies for commercial equipment.

a. Tow vehicles

1. Braking system. Braking system will be functionally compatible with the towed vehicle braking system. The towing vehicle should not add to yaw, sway, skid, tip, tilt, or jack-knife of the towed vehicle under maximum braking conditions and minimum stopping distances.

2. Controls. Controls must be designed to ensure towed vehicles are under positive control.

3. Parking brakes. The parking brakes, together with the towed vehicle parking brakes, will hold a fully loaded towing and towed vehicle combination on an 11.5 degree incline with the tow vehicle headed up or down.

4. Vehicle connecting device. The vehicle connecting device will be compatible with that of the towed vehicle.

5. Fifth wheel. The fifth wheel (if used) will be equipped with a safety latch. The safety latch will be designed to allow a visual check of the locked condition.

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6. Self-centering controls. All movement controls (except for such devices as the parking brake, steering control, transmission selectors, power takeoff, and hydraulic pump) will be self-centering. The engine start switch will operate only in the automatic transmission "neutral" or "park" position.

7. Creep movement. A capability for small increments of forward and reverse movement is required.

b. Transport vehicles. A vehicle used to transport nuclear weapons on its basic structure will satisfy the criteria in sections 5.3.2 through 5.3.3 and have parking brakes that will hold the vehicle with rated load on an 11.5 degree incline with the vehicle headed up or down.

5.3.6 Handling equipment. This section applies to equipment used to lift and load nuclear weapons (such as forklifts, missile lift trucks, high-lift trucks, munition handling trailers with lifting devices, and loading/unloading trucks). The criteria in sections 5.3.2 and 5.3.3 apply, as well as the following:

a. Fail-safe. The equipment will be designed to maintain safe control of the rated load if electrical, hydraulic, or pneumatic system failure occurs.

b. Parallel systems. If more than one power operating component in a mechanically parallel system is used to lift the weapon, the components may be individually controlled to provide weapon attitude adjustments. However, the components will be capable of synchronization to provide a uniformly controlled lifting attitude.

c. Self-centering controls. All movement controls (except for such devices as the parking brake, steering control, transmission selectors, power takeoff, and hydraulic pump) will be self-centering. The engine start switch will operate only when the automatic transmission is in the "neutral" or "park" position.

d. Positive control. Design will provide for positive control of the nuclear weapon at all times in the lifting and handling modes. One control method is to use load attachment points and straps.

e. Motion limiting devices. Mechanical stops or electrical switches will be added to prevent overtravel in all directions of the lift control. Failure of a single motion limiting device shall not result in damage to the weapon.

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f. Parking brakes. Forklift parking brakes will be able to hold a forklift with rated load on an 8.5 degree incline in both forward and reverse directions. Weapon loader service and parking brakes will independently hold a fully loaded weapon loader on an 11.5 degree incline with the loader headed up or down.

g. Tines and adapters. The forklift tines and adapters will be designed to safely meet all nuclear weapon operational requirements for the forklift. The forklift center of gravity must be compatible with this requirement.

h. Creep motion capability. Weapon loader design will include a capability for small increments of movement in both reverse and forward directions.

5.3.7 Hoists, cranes, and similar devices. The following criteria apply to hoists, cranes, winches, and similar equipment. This equipment will be designed, as a minimum, to have:

a. Positive control. Controls to ensure that the load is under positive operator control.

b. Fail-safe stops. Features to make sure prompt, automatic stops are possible if the operating mechanism fails.

c. Automatic stops. Features to automatically stop the device in the absence of operator control.

d. Load and rate limits. Identified limits and rates for maximum lift capacity and positioning.

e. Allowable safety factor. All hoists shall be tested to a minimum of 125 percent and a maximum of 150 percent with a 200 percent static test for 10 minutes at the rated capacity.

f. Hooks. Hooks fitted with throat-opening safety devices.

g. Stops. Stops or limit switches to prevent overtravel of the hoist on rails and to stop the chain or wire rope when the hook reaches its upper limit. Failure of a single motion limiting device shall not result in damage to the weapon.

5.3.8 Commercial vehicles. Commercial equipment shall meet applicable requirements of this document.

5.3.9 Containers. Containers used for storage and transportation of nuclear weapons will incorporate design criteria

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specified in Q-STD-328. Tiedown points must be adequate to enable tieing down the container to meet ground, shipboard and aircraft cargo restraint criteria.

5.3.10 Pallets. Pallets designed for transportation and storage of nuclear weapons will meet the following criteria:

a. Pallet design. Pallet design shall conform to MIL-STD-1366, and MIL-STD-1367. MIL-STD-1660 covers the design criteria for assemblage of items on a pallet.

b. Tiedown rings. Pallets must have sufficient tiedown rings to carry design loads.

5.3.11 Shipboard transportation systems. Transportation and support equipment (including containers and pallets) designed for use onboard ships and small craft will meet the requirements of sections 5.3.1 through 5.3.11. In addition, the equipment will be compatible with shipboard tiedown configurations which shall meet the following criteria:

a. Load criteria. The number and design of equipment tiedown points must be sufficient to restrain design loads based on maximum expected sea state conditions.

b. Separate terminator. Each tiedown device will terminate at a separate attachment point without passing through any other attachment point.

5.3.12 Underway replenishment systems. Shipboard equipment used for the Underway Replenishment (UNREP) of nuclear weapons or nuclear weapons components in accordance with NWP 14-1 shall be designed and tested to preclude damage to nuclear components.

a. Pallets. Pallets used for UNREP of nuclear weapons shall meet performance criteria of MIL-STD-1660.

b. Containers. The containers must meet the shock isolation criteria in MIL-STD-648.

c. Floatation. Sufficient buoyancy shall be considered as part of nuclear weapon UNREP equipment to provide floatation for the nuclear weapon, handling equipment, cables, pallets and other equipment that may be lost overboard in the event of an underway replenishment accident or mishap.

d. Load capacity. Pads, cables, winches, pulleys and other components of underway replenishment equipment shall be designed

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to withstand the maximum expected load with a static safety factor of 5, 4 for a tension system and a dynamic load of 3.

5.3.13 Air transportation systems. Air transportation and support equipment (including containers and pallets) will be designed to meet the general specifications of MIL-A-8421.

a. Tiedown configurations. In addition to meeting the requirements of MIL-A-8421, the nuclear cargo tiedown configurations will:

1. Not impose a reaction load that exceeds the strength of aircraft tiedown rings identified in the applicable cargo aircraft specifications.

2. Limit one tiedown device to one aircraft tiedown ring.

3. Consist of tiedown patterns symmetrical about the longitudinal axis of the aircraft.

4. Require at least four tiedown devices for each item. Only tiedown devices that meet MIL-T-25959 for chains and MIL-T-27260 for straps will be used with nuclear cargo.

5. Require each end of each tiedown device to terminate at a separate attachment point without going through any other attachment point.

b. Pallet configuration. In addition to meeting the requirements of MIL-A-8421 and the requirements of section 5.3.13.a, the nuclear cargo tiedown configuations on pallets will:

1. Meet restrictions defined in the specifications for the pallet.

2. Not exceed allowable deck roller loads identified in the applicable cargo aircraft specifications. The pallets should bridge the rollers satisfactorily.

c. Cargo restraint criteria. The design loads for the tiedown patterns will be determined by applying the minimum Acceleration Due to Gravity (G) load factors shown in Tables I and II. Tiedown patterns will be structurally analyzed or proof-tested (or both) to make sure they meet the proper G load factors before these patterns are published in the aircraft cargo loading specifications. Off-pallet tiedowns which meet

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the criteria in section 5.3.13.a., may be used to secure pallet loads in cargo aircraft.

TABLE I

Typical Nuclear Weapons Tiedown Configuration G Load Factors for Other than COD Aircraft (identified in STS)

Load Direction	G Load
Forward	10.0
Aft	7.5
Lateral	3.0
Vertical (Down)	4.5
Vertical (Up)	3.7

TABLE II

Typical Nuclear Weapons Tiedown Configuration G Load Factors for COD Aircraft (identified in STS)

Load Direction	G Load
Forward	20.0
Aft	7.5
Lateral	3.0
Vertical (Down)	4.5
Vertical (Up)	10.0

5.3.14 Vertical replenishment criteria. Equipment intended for use in vertical replenishment operations involving nuclear weapons will be designed in accordance with the requirements of NWP 14-1.

5.4 Electrical and Fiberoptic Systems

5.4.1 General. A major portion of any nuclear weapon system is composed of subsystems designed to monitor, select, load, prearm, arm, launch, fire, and release nuclear weapons. These subsystems may depend upon electrical energy, light energy (for fiberoptics), or a combination of these. The design of these subsystems must preclude accidental operation, a single component failure, or an optical or electrical disturbance from performing or degrading critical functions.

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5.4.2 Electromagnetic interference. All electronic and electrical subsystems/equipment in or associated with nuclear weapon systems shall be designed so that undesired responses and emissions are minimized. Appropriate portions of MIL-B-5087, MIL-E-6051, MIL-C-25200, MIL-STD-461, MIL-STD-462, MIL-STD-463, MIL-STD-1385, DOD-STD-1399 (Navy), MIL-HDBK-235, and OP 3565 Vol.2 apply. In addition, the design must ensure that the requirements of the National Security Agency TEMPEST program are met for the electrical, electronic, and fiberoptical equipment which process secure codes (NSAP KAG-30-A/TSEC).

a. General. Wires, switches, cable connectors, junction points, and other system elements shall be designed to prevent undesired radiated and conducted interference or transients.

b. Circuit protection. The nuclear weapon monitor and control circuits, as well as launching and firing circuits, shall be protected from Electromagnetic Interference (EMI).Protection from EMI must be carried through all applicable connectors, junction boxes, and other discontinuities, such as access doors and missile module joints.

5.4.3 Optical interference energy. All fiberoptic subsystems in or associated with nuclear weapon systems shall be designed so that undesired responses and emissions are minimized. Fiber-optic cables, connectors, junctions and other system elements shall be designed to prevent undesired radiated interference, transients, or degradation of subsystem performance when such interference or transients could result in a nuclear hazard, undesired release, launch of a weapon, or undesired firing of the weapon propellant. Nuclear weapon monitor circuits, weapon control circuits, release circuits, and nuclear weapon firing circuits shall be protected from Optical Interference Energy (OIE). Protection from OIE must be carried through all applicable fiberoptic cable connectors, junction boxes, and other discontinuities.

5.4.4 Isolation. Isolation must be provided to prevent select, prearm, arm, launch, or fire functions from occurring during catastrophic events or abnormal environments. The following specific requirements apply:

a. Wire shields. Wire shields shall not be used as current carrying connectors. Shields will be covered by an insulation layer.

b. Electroexplosive circuits. MIL-STD-1512 applies to electroexplosive circuitry affecting critical functions.

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c. Command and control, consent functions. Circuits for nuclear functions shall be physically and electrically isolated from all other circuits and from each other.

d. Prearming and safing circuits. Prearming control circuits and safing circuits must be electrically isolated from all other circuits.

e. Power circuits. Prearming power circuits must be physically and electrically isolated from all other circuits; however, the power and control circuits may coincide at the final switching assembly prior to the nuclear weapon interface. Physical isolation within the final switching assembly shall be maintained to the maximum extent possible.

f. Monitor circuits. Monitor circuits must be electrically isolated from power and control circuits. Monitor functions shall be capable of being performed independent of weapon control functions.

g. Logic circuits. All logic circuit signal leads must be located away from power leads and must be routed to prevent short circuits and minimize inductive and capacitive coupling.

h. Nuclear-nonnuclear functions. For hard-wired systems, electrical functions which are peculiar to the nuclear weapon monitor, control, and release systems must not share an electrical connector with nonnuclear fuctions.

i. Functional operation-warhead circuity isolation. Electrical wiring, circuits, components, etc., employed for functional operation of nuclear weapon guidance, must be electrically and physically isolated from the critical warhead circuitry.

j. Stray power sources. Critical circuits must be isolated from potential sources of stray electrical power.

k. Monitoring-functional circuit isolation. Hard-wired monitoring circuits shall be separated from functional circuits. Monitoring may be accomplished through separate circuits leading to back contacts or separate contacts of explosive switches, S/A devices, barometric switches, relays, etc. (The above design practices that logically apply to fiberoptic circuitry and performance characteristics shall apply to the design of fiberoptics used in or with nuclear weapon systems.)

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5.4.5 Alternating current power. Power circuits must have twisted leads with single point grounding of the return. Structure return shall be avoided to eliminate common mode problems.

5.4.6 Thermal batteries. When thermal batteries are utilized, provisions should be incorporated to discharge the energy if the mission is aborted.

5.4.7 Switching. Normally, the ground side of power circuits will not be switched. To meet two-fault safety criteria, ground switching may be used in addition to hot-side switching.

5.4.8 Design criteria for wiring and cabling. The following criteria shall apply to electrical wiring and cabling. These criteria that logically apply to fiberoptic links shall be utilized in the design of nuclear weapon systems utilizing fiberoptics.

a. Electrical coupling and wiring separation. Wiring and cabling design must minimize coupling and provide for optimum separation within the available wiring space. Cable design must provide for adequate termination of shielded wires.

b. Cable construction. Cable construction must provide for twisting of each signal wire and the individual return wire. Equipment design and cable routing shall minimize circulation currents. Where employed, a signal return common to two or more circuits shall use a common ground reference connection.

c. Vibration. Electrical wiring shall be installed and secured to minimize vibration, minimize the effects of vibration, and prevent chafing.

d. Connector selection. Except for weapon and warhead interface connectors, electrical power wiring must terminate in female connectors at the source outlet.

e. Wiring support. Wiring for critical circuits must be provided with mechanical support which is an integral part of the connector at the entry point into the electrical connector.

f. Fiberoptic links. Fiberoptic links will be designed to preclude entry of stray optical radiation. Cable routing, securing methods, and bundling techniques shall be designed to minimize the probability of damage to the links which could allow entry of stray optical radiation.

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5.4.9 Design criteria for electrical connectors. The following criteria shall apply to electrical connectors and to fiberoptic connectors where applicable:

a. Alignment. Electrical connectors shall be designed to prevent misalignment of connectors, minimize bent pins during mating/demating, and prevent mating of adjacent connectors. Only one wire per pin will be used. Connector pins shall not be used as a terminal or tie-point for multiple connectors. Spare pins shall not be used for mechanical support of in-line components. (Such practices result in an inherent multi-fault condition.)

b. Connectors. All hard-wired electrical connectors associated with nuclear weapon circuits shall conform to MIL-C-38999. Electrical connectors must be isolated from non-electrical connectors. Potting compounds, if used in electrical connectors, must positively preclude reversion.

c. Pin assignment. The assignment of functions to pins within a connector shall ensure that a short circuit occuring as a result of a bent or misalinged pin shall not result in delivery of current to the nuclear weapon or to critical function circuits which exceeds the maximum acceptable level for inert effect on the particular equipment involved.

d. Back shell requirements. Connectors used to carry shielded wire shall use a back shell that provides for peripheral bonding of the shield. Nonconductive finishes which would limit the bonding of the shields shall not be used on these connectors.

5.4.10 Electrical current considerations. Monitoring and testing current must be limited to a functional value below the level required to actuate the most sensitive component in the nuclear weapon system.

5.4.11 Design criteria for panel construction. Panel construction for critical electrical functions, if not hermetically sealed, must have all terminals fully insulated (or the panel fully enclosed) so that foreign objects cannot enter to cause terminal-to-terminal or terminal-to-ground short circuits. If a control panel applies power to the nuclear weapon, and is not hermetically sealed, it must be fully enclosed and all electrical contacts insulated. (Drain holes required by applicable military specifications are permissable.) Logic elements used for the control, monitor, and release of nuclear weapons must be fully insulated if not hermetically sealed, to prevent terminal-to-terminal or terminal-to-ground short circuits caused by moisture or foreign objects.

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5.4.12 Electromagnetic radiation hazards to nuclear weapon systems. Nuclear weapon systems shall be designed to protect against the inherent susceptibility of Electroexplosive Devices (EEDs) and semiconductor devices, fiberoptic devices, and other devices to Electromagnetic Radiation (EMR). The systems shall be designed and tested to assure their safety and reliability. As a minimum, the EMR shields specified in MIL-STD-1385 shall be certified as per NAVSEAINST 8020.7.

5.4.13 Design to eliminate or minimize electromagnetic radiation effects. Maximum practicable protection is required against the hazards of EMR, including lightning and electromagnetic pulse. The shielding design philosophy employed must recognize that the entry of electromagnetic energy into a system can occur through cable and wire couplings, windows and deliberate antennas, and the weapon system (acting as an antenna).

5.4.14 Shielding design. The following rules for achieving satisfactory shielding design apply:

a. Place all susceptible components, subsystems, and systems in metallic enclosures (for example, storage and transport containers for missiles, shells, and reentry vehicles provide considerable protection against electromagnetic energy).

b. Provide Radio Frequency (RF) shielding for openings which provide access to cables, connectors, and other electrical equipment.

c. Provide RF shielding for cables attached to susceptible components and for all cables attached externally to the nuclear weapon.

d. Provide attenuators, filter networks, lightning arrestor connectors, or electrical surge arrestors where practicable.

e. Use circuit-isolation relays or switches which are insensitive to RF energy and position them as close as possible to the elements to be protected to prevent energy pickup by long leads.

f. Ensure all section joints of the weapon are low-resistance contacts so that the weapon skin serves as an RF shield.

g. Firing circuits for EED shall consist of two wires, twisted together and shielded. A single ground point shall be common to all firing circuits.

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h. The system shall be designed such that, regardless of the no-fire threshold of the EED, the maximum root mean square current experienced in the bridgewire shall be 20 db below the maximum no-fire current when exposed to the stray electrical energy environment specified in the STS or 100 watts/meter^2 average whichever is greater. However, maximum no-fire current is not the only design criteria of shielding. MIL-STD-1385 has other EMR field requirements that must be met.

i. Monitor circuits shall be designed so that they will not conduct or couple EMR energy into functional or control circuits. A 100 milliampere limitation applies for nuclear weapon/warhead monitoring.

j. The case of an initiator shall be electrically bonded to the structure. All removable parts of the case, which are removed and replaced during operational and maintenance procedures, must not degrade the shielding after replacement or reassembly.

k. Maintain electrical continuity and isolation of circuitry shielding. Shields shall not be used as intentional current carrying conductors. Metallic (water-proofed) shielding caps shall be provided and shall ensure electrical continuity from shield to case with no gaps or discontinuities in the shielding configuration. When EED male connectors are in an open circuited configuration, shielding caps shall be installed.

5.4.15 Electromagnetic radiation environment throughout the stockpile-to-target sequence. A comprehensive survey of the EMR environment which weapons will encounter is not available to date. However, a general conclusion drawn from available measurements is that the ground-based, shipboard, and airborne radio transmitters now in the Navy inventory, generate average field intensities ranging from a few microvolts/meter to about 300 volts/meter, and that radars can generate peak power densities of about 75 dbm/m^2 in the near vicinity of the antenna. The above mentioned power densities are presented here as an indication of the order of magnitude of the EMR environments anticipated. The environments actually experienced by a particular weapon throughout the STS will be determined by the application of well-known antenna theory, once the location and characteristic of all RF emitters in the vicinity of the weapon have been determined.

5.5 Arming and Fuzing Systems

5.5.1 General. The Arming and Fuzing (A&F) system is the sum of component devices, and design features which cause weapon

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prearming, arming, fuzing, or firing, as well as those components and features which protect against deliberate unauthorized or accidental prearming, arming, fuzing, or firing. The criteria set forth in this section are intended to be used by the NAVSEA PMs and their contracted designers and evaluators. Both DOD and DOE subsystems are normally a part of the total weapon system A&F design. Many of the requirements in this section are satisfied by the DOE design. Effective design should incorporate the components and design concepts described herein to satisfy the criteria in paragraph 4.2.

5.5.2 Design standards. Weapon system design will ensure against unauthorized or accidental selection, prearming, and arming signals being provided to the nuclear warhead in normal or abnormal environments. The design will also ensure against premature launch, or firing. The normal and abnormal environment nuclear safety protective features in the warhead can provide maximum protection only if these signals and events are controlled.

5.5.3 Stimuli for arming and fuzing. The A&F system controls the energy required to cause prearming in all warheads, propulsion units and propellant charges. Some systems require a unique authorization stimulus. Missiles, projectiles, and warheads require unique prearming and unique environment or trajectory stimuli for operation. In addition, weapons launched from ships, aircraft, and other vehicles, sense launch or firing from the delivery platform.

5.5.4 Responsibility for stimuli control. The A&F system provides the capability to respond to the unique prearming and environment for trajectory stimuli. NAVSEA is responsible for stimuli control and for protecting against premature transmission of the authorization, prearming, launch or release, and environment or trajectory inputs in both normal and abnormal environments.

5.5.5 Safety design concept. Safety devices and features utilized in achieving nuclear safety will be designed to withstand abnormal environments for a longer time than the critical arming components or until the weapon is rendered physically incapable of providing a nuclear detonation. The DOE nuclear weapon design concept, which has the following characteristics, shall be considered in the design of the NAVSEA/PMs developed parts of A&F systems.

a. Exclusion region. This region contains the firing set and weapon detonator system. It also has the necessary packaging

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and safety devices to exclude electrical energy from the firing set and weapon detonator system, except when their use is intended.

b. Strong links. Access to the firing set for the arming and firing signals is provided through safety devices (such as system prearm devices and environmental or trajectory sensing devices) called strong links. These strong links are designed to provide energy isolation in an abnormal environment and to operate in their normal mode only in a situation when their use is intended.

c. Weak links. A weak link is a selected functional unit (such as a capacitor or transformer) that is vital to the operation of the firing set and weapon detonator system and whose function is not likely to be duplicated or bypassed. Weak links are designed to respond predictably to certain levels and types of abnormal environments by becoming irreversibly inoperative (thus rendering the system inoperable) at levels less than those at which the strong link fails to keep (electrical) isolation. The weak link and strong link are co-located in order to experience essentially the same environment at the same time. (Weak links may be elements of electrical systems, mechanical systems, fiberoptic systems, etc.).

5.5.6 Devices and systems criteria for arming and fuzing. The following criteria apply to shipboard arming and fuzing systems devices and to systems design features:

a. Warhead identification. When nuclear weapons of one type are mixed with nuclear weapons of a different type, or with conventional weapons, the launch platform weapon system shall have the capability to query each nuclear warhead section and receive a fail-safe, unambiguous identification signal, as a result of the query, from the nuclear warhead section. A secondary means of identification will be provided (such as visual, radiographic sensor, etc.). In the event of an incorrect warhead identification, interlocks will prevent any additional movement, or selection, prearming, arming, launching or firing of the weapon, until positive identification is confirmed through secondary means.

b. Prearm device. The A&F system will contain a prearm device located in the warhead, missile, torpedo, projectile, or mine. It will be a strong link safety device that is operated by a Unique Prearming Signal (UPS). This signal will be derived from some part of the weapon system that is under direct human control. If operational considerations permit, the function

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provided by human action will be reversible up to the actual launch, firing, or release of the weapon.

c. UPS signal.

1. General. Inadvertent or accidental application of the prearming signal to the weapon system in normal and abnormal environments will be precluded by proper system design. The design may incorporate the use of a UPS which originates at the man-machine interface of the weapon system where humans enter the UPS, demonstrating human intent (whether authorized or not) to use the nuclear weapon. The intent strong link unique signal is derived from the UPS entered into the weapon system upon intent to use. A basic safety premise is that the entry of the UPS into the weapon system results in a potential loss of the assured safety which is provided by the weapon system in abnormal environments. The UPS is a specific, nonchangeable sequence of bits of a pseudo-random nature specified by the DOE laboratory which contains sufficient information to assure that inadvertent or accidental generation, (e.g., as a result of circuit faults in abnormal environments), is precluded.

2. Implementation concept. The basic implementation concept for UPS entry into the weapon system is that: (1) all of the necessary information be physically separated from the weapon system and retained in such a manner that it is readily apparent if the data has been entered, (intentionally or inadvertently, whether authorized or not), and (2) that the data cannot be accidently inserted, entered, or generated as a result of situations arising from a credible combination of abnormal environments.

3. Implemented considerations. To implement the UPS data entry, the following safety-related design criteria should be met within the weapon system. The fundamental intent of these criteria is to isolate key safety critical components from significant electrical energy in the absence of initiating human action. In the absence of human intent, the UPS must be precluded from the weapon system.

(a) Prevention of accidental UPS entry. All of the information necessary to generate the UPS should be isolated from the weapon system, (e.g., physically separated), until entry is intended. There should be no prior storage of the UPS data, or representation of the data such that it could be accidently entered into or generated by the weapon system. Word lengths and information content for UPS will be predicated on an analysis which assumes a single try implementation. Algorithms

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which detect errors in the UPS data and automatically retry if not correct, should not be incorporated into design if they may create the possibility of multi-try capability in abnormal environments.

(b) Prevention of inadvertent UPS entry. The data entry method should provide positive design features to prevent inadvertent human actions which could result in an insertion of the UPS data. These include mechanical features to assure that inadvertent operation or normal operation for non-nuclear functions will not cause UPS data insertion.

(c) Positive erasure. The design of the UPS data entry system should assure positive and verifiable erasure of the UPS data after use in the system during system tests involving training or test rounds, or after use authorization of the nuclear weapon has been withdrawn.

4. Implementation alternatives. Two general classes of UPS data entry methods are compatible with these considerations. The first is one in which the human operator physically inserts the UPS data into the weapon system with a device such as a read only memory plug, "credit card", magnetic tape, or other physical means. This type of UPS data entry system offers the most straight forward implementation of the stated considerations. The second method uses an input device such as a keyboard for the operator to enter the UPS data. If this design approach is used, abnormal environment safety considerations may dictate that more keyboard strokes be used to enter the UPS data than would be needed to satisfy data requirements, since significant fragments of the UPS may be generated with a single keystroke.

d. Environment sensor. The A&F system will contain an Environment or Trajectory Sensing Device (E/TSD), (in appropriate nuclear weapon system), preferably in the warhead or projectile. The E/TSD is a strong link device that will prevent arming until the proper environment is sensed. This device will only respond to an environment unique to the terminal employment of the weapon. Because this stimulus may occur as a result of the terminal employment of the weapon, it is essential that premature employment or insertion of the weapon into its terminal environment be prevented.

e. Launch or release sensing device. The A&F system will contain a device that will prevent power from being applied to the arming and fuzing system. For ground or ship launched weapons, this device (such as a pressure actuated valve or an impulse sensing switch), will sense the proper environment unique

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to launch. Other types of weapons, such as mines, may utilize other sensors which would have a unique response to environments characteristic to the employment of that type of weapon. These devices will be designed to ensure against accidental or inadvertent operations.

f. Abnormal environment protection. Abnormal environment protection features will prevent prearming and arming in all abnormal environments specified in the STS document and in the applicable carrier vehicle specification.

g. Dual signal arming. At least two separate and independently derived signals, which cannot be generated by a single signal at any point, will be required to arm the weapon. These signals will be interrupted by one or more strong link devices. At least one of these signals will be continuous after application. This is not a requirement for multiple power sources. (After the A&F system has been prearmed, but before insertion of the weapon into its terminal environment, an A&F system failure shall not result in arming the weapon).

h. Energy discharge. When applicable, the A&F system design will provide for automatic discharge of stored energy in the A&F energy storage devices (such as capacitors and activated batteries) if arming power is interrupted. The A&F system will not automatically return to an armed status upon the unintentional restoration of interrupted power, without the proper resequencing of the weapon prearm and arming procedures.

i. Lightning protection. Lightning protection will be incorporated to protect critical A&F circuits. Reference shall be made to the STS document.

j. Cable and connector design. Cable and connector design and connector pin assignment will protect against premature transmission of prearm and arming power to the warhead or the weapon as the result of damaged cables and connectors. The design will guard against cable or connector selection and cable routing that is susceptible to damage during assembly, maintenance and test operations.

k. Nondestructive testing compatibility. Nuclear safety for the A&F system will not be degraded as a result of exposure to standard Navy nondestructive testing environments specified in use for the weapon systems. These tests include, but are not limited to, X-ray, ultrasonic, magnetic, and similar tests.

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l. System monitoring requirements for A&F. The capacity to monitor the state of at least one strong link may be provided in all weapon system configurations. The extent to which this capacity will be used is system dependent. In some systems this monitoring capability may only be used in emergency situations to determine the safe status of the weapon. In others, monitoring may be required periodically during alert operations. While the requirements to provide this capability are always placed on the A&F systems, the requirements placed on the weapon system to use the capability will be developed to complement the employment concept. The monitoring function design will prohibit the possibility of introducing energy from any source that might operate an A&F critical function if a system fault or abnormal environment occurs. Non-electrical monitor systems should be considered.

m. Chemical compatibility and reversion. All material used in the design will be chemically compatible in stockpile and storage environments. No materials will be used that could increase the high explosive sensitivity, generate an explosive or toxic gas, cause an electrical short, cause reversion, or similar results.

n. Input and output isolation. Signal inputs, such as electrical current, fiberoptic light energy, or other similar energy to nuclear safety devices will be isolated from the outputs. Other methods, (such as incompatible signals), will also be used to minimize the possibility of bypassing the safety devices.

o. Dual enable system. Systems which contain integrated real-time processors to determine fuzing parameters would be designed to have two independent systems to enable fuzing. The interplay of the two systems will be controlled in the same manner as dual signal arming requirements discussed in paragraph 5.5.6.e.

5.6 Nuclear Weapon System Automata

5.6.1 General. These design criteria apply to automata and software which receive, store, process, or transmit data to select, load, prearm, arm, enable, unlock, fire, or launch a nuclear weapon. Such equipment shall be designed to provide the highest degree of protection against accidental or unauthorized activation of critical command and control functions. General guidelines for weapon system software development are provided in DOD-STD-1679 (Navy).

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5.6.2 Memory characteristics. Memory characteristics will ensure that contents of memory are not altered nor degraded over time during operational use. Memories with volatile characteristics will have provisions to ensure that erroneous or unexpected information from them, due to power removal or other phenomena that change their state, will not affect any critical function or component. The system will inhibit any memory changes due to hardware failure. A single hardware fault shall not cause a memory change that could initiate a critical function.

5.6.3 Program loading. The system shall be designed to prevent automatic control until all valid and correct program data have been loaded and verified. It will also be capable of detecting errors that may occur during loading. Upon error detection, the system shall halt loading and prevent the use of the data.

5.6.4 Memory program and data change. Once the memory has been loaded and verified, the program and data shall be protected against unauthorized changes.

a. Hardware faults or failures. Memory changes due to hardware failures or faults shall be minimized to the maximum extent possible. A single hardware fault or failure must not result in a memory change involving critical functions.

b. Reloading or changing memory. Reloading or changing the memory critical functions or contents shall require a method for proper entry to the memory. An attempt to gain access to memory or to load or change it without controlled means for entry shall cause a positive indication to the weapon system operator and/or maintenance crew, and the entry will be automatically rejected.

c. Program transfer. If selected computer software routines are stored at locations remote to the nuclear weapon system, and are to be transferred by communication systems, a method will be devised to ensure that the routines have been transferred correctly. A method must be devised to prevent unauthorized or inadvertent transfers.

5.6.5 Processor control. Automata which contain nuclear weapon system critical logic will be designed so that deviation from proper routine or sequence will automatically result in a recycle to the proper sequence or a self-test mode. If this requirement cannot be met, an automatic shut down with an associated signal to the operator will be provided.

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a. Operator control. In operational, maintenance, or failure modes, automata that control or monitor nuclear weapons will not be able to bypass the operator control of critical functions.

b. Program control. If automata are entirely program controlled, safety features shall be incorporated to prevent improper sequences or routines from occuring.

c. Improper processing. Detection of improper processing shall be provided in all cases where continuation of automatic program control would degrade nuclear safety.

5.6.6 Program design. System requirements will be translated into program design in a systematic method. The method will be hierarchical structure of identifiable programs, subprograms, modules, procedures, and routines. The highest level of control logic will reside at the top levels, and computational or algorithmic functions will reside at lower levels. Levels will be set up such that lower levels do not call on higher levels. Program structuring techniques will be used to the maximum extent consistent with the programming language selected.

5.6.7 Hardware cross-check. The design of automata for critical functions shall provide self-check, confidence, or test routines. Such automata shall have a first or second level interface (section 5.6.8) with a nuclear weapon and thus, if tampering has occurred, an unauthorized action fault shall be signalled. Automata design should incorporate methods for cross-check, such as:

a. Special test circuitry. This circuitry is designed to self-test the hardware and disclose tampering.

b. Certification and confidence programs. Software programs may be specially developed to check critical circuits. Software checks must be designed so that routines within the programs will indicate the possibilities of unauthorized action.

5.6.8 Software nuclear safety analysis. Any software used to process the data for, provide the status of, or exercise automated/automatic control over any critical nuclear weapon system function, shall be subjected to a Software Nuclear Safety Analysis (SNSA) to ensure system nuclear safety integrity.

a. Purpose and requirements. The SNSA provides assurances that the software or other program controls as designed, coded, and implemented, cannot contribute to accidental, unauthorized, or fault activation of critical nuclear weapon system functions.

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b. Guidance. Software components will be designated for a SNSA based upon an analysis of the degree to which the software could affect critical functions, or contribute to an unauthorized launch or release. Software is categorized by degree of involvement as follows:

1. First level interface. Any software used by automata having a direct electrical connection to a nuclear weapon is categorized as a first level interface. This includes all resident or processor accessible programs, even though no critical functions are performed.

2. Second level interface. Any software used by automata having a direct electrical connection to automata having a first level interface, is categorized as a second level interface. This includes all resident or processor accessible programs even though no critical functions are performed.

c. Software currency requirements. Nuclear safety critical software analyses shall be conducted throughout the weapons system life cycle whenever new or modified software is developed. Modified software may have a modified SNSA covering the changed segments, but all interfaces to the original program must be checked. Software which is not subject to recurring changes, requires only an initial review, analysis, and certification.

d. Special diagnostic software. Certain software for maintenance or diagnostic purposes may require an SNSA.

1. Diagnostic software planned for use must receive an SNSA if the appropriate computer memory overwrite is not planned to be loaded before the diagnostic tape.

2. Diagnostic software planned for use must receive an SNSA if cryptographic codes, variables, and routines in the computer memory are not planned to be overwritten or erased before the diagnostic software is loaded.

3. Positive measures must be taken to preclude the possibility of residual diagnostic software instructions remaining in memory after completion of any maintenance or diagnostic tests.

5.6.9 Development of software nuclear safety analysis. The organization performing the SNSA, and the software developer organization, must be independent of each other. The PM has the responsibility for coordinating the liaison between these two organizations and for qualifying such analytical tools as may be

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developed to perform the SNSA. Generally, these tools shall not have been developed by the software developer organization. The integrity of software analysis tools must be assured through proper security and/or control procedures. The SNSA shall be conducted and documented to verify that:

a. The software to be checked is identical to the engineering original or master.

b. Critical areas, routines, program branches, switches and commands within the software are identified and checked.

c. No unauthorized programming is present that could affect performance of nuclear safety critical functions.

d. Under all specified input conditions, the program response is completely consistent with nuclear safety design and evaluation criteria.

e. A high level of confidence is achieved under all operational and maintenance conditions that the program will not cause or contribute to the improper performance of the critical functions which affect the safety of the nuclear weapon system.

5.6.10 Mechanical design criteria. Automata hardware and that hardware controlled by automata equipment will be designed to preclude deliberate or unauthorized actuation of any nuclear weapon or nuclear weapon system critical function. Degradation of automata, hardware or hardware controlled by automata due to environmental effects, shall not permit the bypassing of the required sequencing or critical functions, nor allow deliberate or unauthorized alteration of firmware or software. The design and test environments shall be specified by the procuring activity.

5.7 Test and Training Equipment

5.7.1 General. The design criteria guidelines in this section apply to test equipment and training devices for nuclear weapon systems. These guidelines are not necessarily all inclusive and should not limit the designer from introducing new, innovative ideas to further improve nuclear safety. War reserve weapons shall never be used as a test device or as a training device. Where several agencies provide components of a system, coordination of the design is mandatory to assure establishment and maintenance of a safe interface. In the event of conflicting requirements, the procuring activity shall determine the guiding requirement.

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5.7.2 Test equipment. Equipment used to test nuclear weapon systems shall meet the following design criteria:

a. Utilization. Test equipment shall be used only where necessary to establish and verify system operation, reliability, and safety. The majority of any required testing shall be done before the nuclear weapon is mated to the combat delivery vehicle. Testing should be minimized after a nuclear weapon is mated to its delivery system and/or delivered to its organizational level unit.

b. Test intervals. Intervals between required tests on nuclear weapons or weapon systems shall be the maximum interval permissible to maintain a high confidence level in the system operation and safety.

c. Malfunction isolation. The design should ensure that no malfunction or fault in the test equipment or test circuits can occur which could operate nuclear weapon critical functions or which could degrade the nuclear safety of the equipment being tested.

d. Electrical isolation. The design should ensure that the introduction of stray voltages, unlimited current, or EMR energy into the weapon system is minimized. Interference and susceptibility within test equipment shall be controlled by adequate provisions to eliminate undesired responses and emissions from all electronic and electrical equipment in, or associated with, nuclear weapon systems.

e. Safing. Test equipment must ensure that any weapon system component which has been operated during testing is in the safe or unactivated condition at the completion of the test. The safe condition of such components shall be verified by a positive indication.

f. Fault identification. Based on system requirements, test equipment shall reveal improper wiring, line-to-line shorts, line-to-ground shorts, stray or improper voltages, and improper system operation.

g. Test equipment stability. When nuclear weapons are present, electrical test equipment, including built-in test equipment, must be incapable of:

1. Generating firing or launch signals.

2. Causing any critical nuclear function to occur.

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3. Negating two-man control functions.

4. Unlocking any safety interlock on the launch vehicle

5. Activating an Arm/Disarm or S/A device.

h. Automata. Automata and software used for testing or checkout must meet the provisions of section 5.6 of this handbook.

i. Power limitations. Testers shall not be capable of applying sufficient power to cause the operation or firing of any item being tested except when specifically designed to do so. Electric currents for monitoring must be limited to values below the level capable of functioning the most sensitive component of the warhead section.

j. Junctions, switches, and connectors. Wires, switches, cable connectors, junction points, and other electrical system elements, as appropriate, shall be designed to prevent undesired radiated and conducted interferences or transients.

k. Monitoring currents. Monitoring currents should be placed on non-functional circuits through separate circuits leading to separate contacts of relays, electroexplosive devices, safe/arm devices, and barometric switches.

l. State of the art methods. Wherever possible, the use of blocking schemes, fiberoptics, and any other state of the art system(s) to further isolate and protect critical circuits from those under test are encouraged.

5.7.3 Devices and simulators. Training devices or nuclear weapons simulators shall not be used with nuclear weapons. Further, training devices and simulators shall not be capable of arming or launching a nuclear warhead. Those items used to duplicate a nuclear weapon or weapon system function, must have an explicit means of identification as a training item.

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6. NUCLEAR WEAPON SAFETY EVALUATION CRITERIA

6.1 Nuclear Weapon Systems

6.1.1 Shipboard. Evaluation requirements and procedures for obtaining nuclear safety approval of a shipboard nuclear weapon consist of an evaluation of the design by analysis and test (where applicable) and a study and physical examination of the weapon system in accordance with OPNAVINST 8110.20. The primary requirements for nuclear safety for shipboard systems are described in the applicable design criteria sections.

6.1.2 Ground delivered. When the implementation of system or equipment specifications will result in hazards, (see MIL-STD-1574), the system manager will conduct a trade-off study to achieve maximum nuclear safety consistent with operational requirements. The primary requirements for nuclear safety for ground delivered systems are described in the applicable design criteria sections.

6.2 Transportation (Logistic and Tactical) and Related Support Equipment

6.2.1 General. Evaluation requirements for certification consists of analysis, examination, or testing by the responsible Navy agency. Weapons involved in logistic transport will not be in the completely assembled for launch configuration unless specifically authorized.

a. Fail-safe features. If fail-safe features are incorporated, they must be evaluated or tested to determine that they provide safe control of the weapon in any system failure. The weapon may not be supported by lift arms, cables, or hydraulic system, but must be supported by the basic frame of the equipment during both air and ground transport. (This requirement does not apply to equipment used solely to position or transfer weapons.)

b. Fire or shock. The equipment must prevent or reduce the transmission of fire or shock to nuclear weapons. This requirement can be demonstrated by analysis, testing, or a combination of both.

c. Roadability. The equipment must meet minimum roadability requirements as specified in the Stockpile-to-Target Sequence (STS).

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d. First article analyses and structural tests. These analyses or tests determine that the design criteria of section 5.3.3 have been met. (For wheeled vehicle testing, the article is to be supported only by the structural members since it is not intended that tires withstand required test load.) Adequacy of the designated structural safety design factor, (a minimum of three times the static rated load, or two times the dynamic load, whichever is larger in each axis), shall be proven by one or more of the following ways:

1. Performance of a detailed stress analysis. This analysis may be supplemented by selective structural test.

2. Nondestructive tests to conditions which approach the yield strength of the structure with suitable instrumentation at all critical stress points as determined by stress analysis. These tests should not cause yielding of any component. (Yield has occurred when strain instrumentation reveals a permanent set, as defined by material properties.) Results of these tests shall be combined with an abbreviated stress analysis to determine whether the design meets the required design criteria. The test design load must be determined by analysis, and must include the static design load or appropriate dynamic design load (whichever is greater in each axis). Mobility test results will be used to determine the total dynamic load. These test loads must be applied simultaneously to the test vehicle along the appropriate axis.

3. Destructive test loads shall be applied simultaneously to the test vehicle along the appropriate axis until the item fails. The test loads at this point must be related to a load at the material yield point greater than the static design load or the dynamic design load.

e. Operational proof tests. Tests shall be performed on at least one fully configured article and other designated samples as necessary to determine that the item will withstand specified rated loads. Example methods are:

1. Testing the first fully configured article to 110 percent of rated load and all other articles to 100 percent of rated load.

2. Testing the first fully configured article and selected samples to 110 percent of rated load.

f. Rated load and gross weight. Rated load and gross weight shall be clearly identified on all equipment.

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6.2.2 Forklifts and weapon loaders.

a. First article and proof load test. The test requirements of sections 6.2.1.d. and 6.2.1.e. shall be met.

b. Temperatures. Test all lifting devices at test site ambient temperatures and then repeat at the extreme operating temperatures using rated load capacity.

c. Drift rate tests. Conduct drift rate tests with the lifting device positioned at extreme and midheights. Internal leakage in hydraulic components of lift systems must be limited so that the maximum drift rate is consistent with the precision required to prevent damage to a weapon.

d. Parallel systems. If more than one power actuating component in a mechanically parallel system is used in lifting the weapon, the component may be individually controlled to provide for weapon attitude adjustments and must be capable of synchronization for a uniformly controlled lifting attitude.

e. Overpressure. Determine that hydraulic and pneumatic systems are designed to prevent overpressure.

f. Attachments or lifting devices. Analyze or test any attachment or lifting device fitted to a vehicle for compatibility in the worst case environment to which the vehicle will be subjected.

g. Parking brakes. Test the parking brakes to ensure that a fully loaded vehicle does not roll on slopes as specified in section 5.3.

6.2.3 Commercial vehicles. If it does not meet the requirements of SWOP H-1, equipment shall be evaluated for nuclear safety adequacy in accordance with appropriate standards, specifications, designated tests, etc. Commercial vehicles must meet the requirements of this document to the extent of its use with nuclear weapons. The responsible evaluation agency must ensure that nuclear safety is not compromised for the intended use.

6.2.4 Trailers, tow vehicles and self-propelled vehicles. Test the service brakes of all vehicles, including tow vehicles and towed vehicle combination brake systems, as applicable.

a. Service brakes. Test the service brakes of all vehicles by progressively increasing the speed from which stops are made, in increments of approximately 5 mph, up to the maximum rated speed of the vehicle, until failure occurs, (or until the maximum

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safe speed has been attained, whichever occurs first). Make the initial tests on a dry, brushed, level, concrete surface. Stop the vehicles by operating the brake system to produce maximum braking force (that is, "panic stop" conditions). Repeat the procedure on surfaces similar to those expected during the operational life of the vehicle. In each test, determine the maximum safe towing speed and record the following data on the brake performance as a minimum:

1. Tendency to yaw, sway or skid.

2. Tendency to tilt, turn over or jackknife.

3. Damage or excessive wear.

4. Stopping distances.

5. Towing speed.

6. Contact of wheels with ground.

b. Accidental trailer towbar disengagement. Test the emergency braking system of trailers using towbars. Accidental trailer towbar disengagement may be verified by full scale testing or by analysis and limited testing. Limited testing may be specified by the procurement/development agency and nuclear safety evaluation agency. For full scale testing, the following will be accomplished: While towing the fully loaded trailer at maximum expected operating speeds over a straight, smoothly paved road, disengage the towbar from the tow vehicle, and watch the emergency braking action of the trailer. The following items must be considered to determine whether the emergency brake system is adequate:

1. Distance from point of towbar disengagement to final stop.

2. Lateral distance of travel from point of towbar disengagement to final stop.

3. Attitude of trailer at time of stop.

4. Damage incurred by trailer and/or load as a result of disengagement.

c. Parking brakes. Test or analyze the parking brakes to ensure that the fully loaded vehicle does not roll when parked on an 11.5 degree slope. Conduct the test with both the forward

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and aft end of the trailer pointed up the slope. The procurement/development agency and nuclear safety evaluation agency will specify if analysis is to be used in lieu of testing.

d. Associated equipment. Test all equipment used to load nuclear weapons on vehicle basic structure against the criteria of section 6.2.1.d. and 6.2.1.e.

e. Mobility requirements. The procurement/development agency and the nuclear safety evaluation agency shall determine the applicable mobility requirements of MIL-M-8090.

f. Ground transport tiedowns. Ensure that nuclear weapon ground transportation equipment is compatible with tiedown techniques specified in section 5.3.2.d.

6.2.5 Combat ships and shipboard transportation systems. Shipboard equipment, devices, subsystems, systems, magazines, nuclear weapon handling equipment, elevators, loading devices, launchers, etc., shall be evaluated to the extent of their intended use with respect to nuclear weapon and nuclear weapon systems safety in accordance with the requirements that are set forth herein.

a. Structural evaluation. Shipboard compartment and magazine structure, tiedown points, and other integrally constructed hardware shall be evaluated by structural test, structural analysis, or a combination of both to demonstrate the adequacy to withstand the design loads required to achieve nuclear safety.

b. Electrical and electronics hazards. Electrical and electronic computers and controls, circuits, circuit breakers, switches, transformers, etc., that are integral to the ship's equipment, shall be evaluated for proper grounding, Electromagnetic Radiation (EMR) and Electromagnetic Interference (EMI) hazards, and other electrical or electronic hazards with respect to nuclear safety to the extent of their intended use.

c. Shipboard safety devices. Shipboard sprinklers, fire and smoke detection devices and other shipboard fire fighting equipment, shall be evaluated with respect to the requirements of this document in so far as nuclear safety is required.

d. Safety interlocks test. The system will be evaluated to ensure proper functioning of safety interlocks intended to preclude weapon or platform damage while the weapon is being loaded, aimed, launched, or fired.

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e. Underway replenishment. Shipboard equipment that is utilized in the underway replenishment of nuclear weapons shall be evaluated with respect to requirements in section 5.3.12. Tackle, winches, booms, padeyes, etc., shall be evaluated by use of structural analyses or properly designed proof testing. NWP-14 shall be reviewed to ensure compliance.

6.2.6 Cargo aircraft compatibility. All equipment used for the transport of nuclear weapons in cargo aircraft shall be evaluated with respect to nuclear safety.

a. Tiedown strength. Tiedowns must satisfy design critieria in section 5.3.2.

b. Tiedown configuration. Tiedown patterns shall be structurally tested or verified by analysis according to load factors of section 5.3.13. Pallet loads, when tested, will be secured to a simulated aircraft section.

c. Tiedown test or analyses. Simulated loads shall be applied according to Tables I and II in section 5.3.13 in the development of tiedown patterns. A tiedown pattern is considered acceptable if the rated strengths of the aircraft floor and pallet fittings are not exceeded, the rated strengths of the restraint devices are not exceeded, and the total load on any tiedown devices attached to the test item does not exceed its rated strength. During application of simulated loads, a determination shall be made on required use of additional aids (such as under the axle or under the frame shoring) for equipment with flexible suspension systems and pneumatic tires.

6.2.7 Reusable containers. The acceptance criteria contained in MIL-STD-648 and Q-STD-328 apply. Verify that tiedown points are adequate to enable tieing down the container to meet ground, shipboard, and aircraft cargo restraint criteria.

6.2.8 Hoists, cranes and similar devices. The procurement/development agency and the nuclear safety evaluation agency shall determine if the following factors have been met:

a. Determine if the design factors have been met.

b. Identify additional tests if necessary.

c. All hoists shall be tested to a minimum of 125 percent and a maximum of 150 percent with a 200 percent static test for 10 minutes at their rated capacity.

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d. Test results at rated load shall be used to determine effectiveness of the fail-safe and synchronization features.

e. Electrical, hydraulic and pneumatic systems shall be analyzed for safe operation at rated capacity.

f. Safety devices (for example, braking devices, locking pawls, safety catches, rail stops, limit switches and valves) shall be tested at rated load.

g. The procurement/development agency and nuclear safety evaluation agency shall determine the required environmental tests.

6.3 Electrical and Fiberoptical Systems

6.3.1 General evaluation of electromagnetic pulse and electromagnetic radiation requirements. Evaluation of electromagnetic pulse and EMR effects must be performed to ensure Electromagnetic Compatability (EMC) of all military communications-electronics equipment, subsystems, and systems during conceptual, design, acquisition, and operational phases. In evaluating the system's vulnerability to credible EMR environments throughout the STS, consider these documents from the DOD EMC program: EMC Program Plan, EMC Control Plan, EMC and EMI Measurement Program Plan, and the EMC/EMI Measurement Test Report. Before the system is certified, the evaluation agency must have the evaluation data that covers EMR hazards to electroexplosive devices, semi-conductors, fiberoptical, and other devices.

6.3.2 Isolation requirements. Tests and analyses will be conducted as necessary to make sure electrical (and optical, if appropriate) isolation requirements have been met in the system design.

6.4 Arming and Fuzing Systems. All Arming and Fuzing (A&F) systems will be evaluated at the system and component levels (through analysis, tests, and demonstrations) to assure nuclear safety during those phases specified in the STS.

6.4.1 Specific data requirements. Specific data will include:

a. Critical components. Identification of each critical component (and its failure modes) whose fault or failure could contribute to premature operation of the A&F system.

b. Failure mode information and information source. Qualitative information on the likelihood of each identified failure mode and the source of such information.

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c. Fault tree analysis. A fault tree analysis of all critical component failure modes that could contribute to premature operation of the arming and fuzing system.

d. Normal and abnormal environments. A nuclear safety analysis of the system in both normal and abnormal environments which, as a minimum, evaluates compliance with the requirements for precluding:

1. Premature nuclear detonation during storage and logistics operations (system not prearmed).

2. Premature nuclear detonation for each stage of prearming and arming.

3. Premature nuclear detonation after the system is armed.

e. Thermal analyses. Thermal analyses with a graphic "temperature-time" presentation showing relative component temperature as a function of exposure time. These analyses will cover every component that can be thermally operated, every thermal-protective component and design feature, and all explosive components.

f. Response characteristics. Graphic presentations of all operate and non-operate response characteristics for components in both normal and abnormal environments.

g. Bent pin and connector misalignment analyses. Bent pin and connector mismating and misalignment analyses for each connector to protect against premature operation of the arming and fuzing system.

h. Cable routing analysis. A cable routing analysis to ensure protection against cable damage throughout the STS which could contribute to premature operation of the arming and fuzing system.

6.4.2 Other requirements. Design criteria in section 5.5 that are not included in at least one of the above requirements, will be demonstrated or analyzed individually.

6.5 Nuclear Weapons System Automata. Nuclear weapon system automata hardware and software which control critical functions will be nuclear safety certified. To ensure that automata and associated software meet the nuclear safety standards, the following nuclear safety design areas will be evaluated:

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6.5.1 Memory characteristics. Memory characteristics will be evaluated by demonstration or analyses to ensure that critical contents are not altered or degraded over time. Demonstrate (or show by analysis) that memory changes in a volatile memory, or a single hardware fault, will not result in a memory change that could initiate a critical function.

6.5.2 Memory stability. Evaluate the following features regarding memory loading and change:

a. Errors are detected and the operators notified.

b. Reset, halt, or synchronization functions are operable.

c. Automatic operation is stopped until all valid and correct data have been loaded and verified.

d. Unauthorized or incorrect loading, reloading or changing of memory involving critical functions is stopped unless the proper means for entry is used. Also that, improper load, reload, or change will be indicated and rejected.

e. Each section of memory utilized is correctly filled by the block of proper program data or instructions to be transferred.

6.5.3 Text erasures. Evaluate the adequacy of the method used to erase memory.

6.5.4 Processor deviations. Processor deviations must be properly evaluated. Evaluate the capability to detect manually started commands, message errors, and deviations causing an erroneous entry to a critical routine. Show that, upon detecting errors and/or deviations, the automata will stop transmitting critical commands and will either recycle, self-test, or perform automatic shutdown while providing an indication to the crew.

6.5.5 Fault detection. Evaluate the fault detection capability of the system.

6.5.6 Hardware evaluation. The following hardware evaluation shall be performed:

a. Self-check, confidence or test routines.

b. Evaluation of the hardware to make sure it meets current National Security Agency TEMPEST requirements (see MIL-STD-461). (This requirement does not apply to functional areas that do not process secure codes).

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6.5.7 Ancillary equipment. Ancillary equipment provides the interface between a human operator and the automata. Ancillary equipment will be evaluated with regard to the extent of control it exerts on the weapon system critical functions.

6.5.8 Hardware and software security. All nuclear weapons system hardware and software components will be evaluated to ensure that the security requirements of DODDIR 5210.41 are met.

6.6 Test and Training Equipment. The criteria in this section are to be applied to nuclear weapon system built-in test equipment, launch or fire control test equipment, nuclear weapon or warhead testers, component testers and general test equipment. The Navy development agency will establish specific criteria for tests, analyses, and demonstrations in the following areas:

6.6.1 Environmental tests. Conduct environmental tests according to the requirements of section 4.4.7.

6.6.2 Analyses

a. Circuit analyses. Circuit analyses of test equipment operating with the circuits of the equipment to be tested.

b. Failure modes and effects analysis. Failure modes and effects analysis of test devices to ensure that faults within the devices will not degrade the nuclear safety of the equipment to be tested.

c. Tester interface analysis. Analysis of the tester interface with the weapon system to verify the test concept.

d. Analysis of maintenance procedures. Provide analysis of procedures for the maintenance and inspection of testers to make sure the integrity of the testers can be verified before they are used.

6.6.3 Demonstrations

a. Fit and function demonstration. Perform a fit and function demonstration to make sure both the mechanical and electrical designs are compatible with the weapon system to be tested.

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b. Demonstration of procedures. Demonstrate operations and procedures. Verify each application section of each procedural document.

Custodian:
Navy - OS

Review Activity:
Navy - OS, SH

Preparing Activity:
Navy - OS

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APPENDIX A

10. SAFETY CRITERIA CHECKLIST

10.1 Scope. This checklist summarizes the safety criteria to be considered by contractors, program managers and others involved in the design, construction, manufacture and operation of nuclear weapons. In all areas, the safety considerations are based on the following four standards set by the DOD as minimum requirements:

a. There shall be positive measures to prevent nuclear weapons involved in accidents or incidents, or jettisoned weapons, from producing a nuclear yield.

b. There shall be positive measures to prevent DELIBERATE prearming, arming, launching, firing, or releasing of nuclear weapons, except upon execution of emergency war orders or when directed by competent authority.

c. There shall be positive measures to prevent INADVERTENT prearming, arming, launching, firing, or releasing of nulcear weapons in all normal and credible abnormal environments.

d. There shall be positive measures to ensure adequate security of nuclear weapons, pursuant to DOD Directive 5210.41.

These four criteria should represent the essence of all safety considerations when involved with any aspect of the nuclear weapon.

10.2 Purpose. The following checklist is a general guide for nuclear weapons and nuclear weapon systems safety design and may not be all encompassing. The designer is responsible for ensuring that all criteria are adequately and properly applied.

20. Weapon System

20.1 The type of weapon system being analyzed, (missile, gun fired, torpedo, etc.).

20.2 Areas of the weapon system being reviewed for safety:

a. Warhead
b. Launcher
c. Handling and transportation
d. Administrative procedures
e. Magazines

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f. Assembly areas
g. Command and control
h. Software processing
i. Other (specify)

20.3 Does this system involve the use of aircraft in the delivery of the weapon in the final leg to the target?

20.4 Are the four basic safety standards of OPNAVINST 8110.20 met? (This question should be answered only at the completion of the safety analyses).

20.5 Are there any deviations from the standard criteria in force?

20.6 Possibility of unauthorized deliberate actions in the following areas must be considered:

a. Selection of a nuclear weapon
b. Prearming
c. Arming
d. Launching, firing or jettison
e. Command disable

30. Weapon Selection

30.1 Describe how the nuclear weapon is selected.

30.2 Can an individual acting alone select a nuclear weapon?

30.3 Once a nuclear weapon is selected, can an individual acting alone prearm the weapon or weapon propulsion system?

30.4 What locations would an individual have to gain access to in order to accomplish No. 30.3?

40. Controls and Switches

40.1 Are the switches, controls and devices for nuclear weapons distinct in shape and distinct in mode of operation from immediately adjacent non-nuclear weapon controls?

40.2 Are switches and controls so designed as to require two distinct movements?

40.3 Describe the controls (mechanical, electrical and administrative) that are utilized in this system to prevent unauthorized activiation of any of the nuclear weapon critical functions.

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40.4 Do switches and controls incorporate devices to prevent inadvertent activation?

40.5 Can any conventional weapon system switch, control or device, when operated in its intended mode, cause any of the nuclear weapon responses listed below?

a. Select
b. Prearm
c. Arm
d. Launch, fire or jettison
e. Detonate

40.6 Can any conventional weapon system switch, control or device, when in a failed mode, cause any of the weapon responses listed below?

a. Select
b. Prearm
c. Arm
d. Launch, fire or jettison
e. Detonate

40.7 Can any nuclear control be activated by using a single motion, such as, pushing a button or turning a knob?

40.8 Are the control switches, knobs, etc., that are used to control nuclear critical functions, distinctly different in shape and mode of operation from any adjacent non-nuclear control switches, knobs or devices?

40.9 Can any nuclear weapon system switch, control or device, when in a failed mode, cause any of the nuclear weapon responses listed below?

a. Select
b. Prearm
c. Arm
d. Launch, fire or jettison
e. Detonate
f. Command disable

40.10 Are the switches, controls, devices and circuitry controlling these critical functions separate and distinct from those that control any conventional weapon common to this weapon system?

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40.11 Are the switches, controls, etc., that control the critical functions of the nuclear weapon, distinct in both shape and mode of operation from any conventioanl weapon switches, controls, etc., located in the proximity?

40.12 Is there any switch, device, etc., in the weapon system that is common to both nuclear weapons and conventional weapons? If so, can any nuclear weapon critical function be activated accidentally or in an unauthorized manner through such a common element?

40.13 Can the operation or activation of any element in a conventional weapon system control, switch, device, etc., cause the activation of any nuclear critical function?

40.14 Are the switches, controls, or devices used to control a nuclear weapon propulsion system distinct and separate from those switches, controls, or devices used to control nuclear weapon warhead critical functions?

50. Power Failure

50.1 Will a power failure result in a nuclear weapon response of the following type:

a. Selection
b. Prearming
c. Arming
d. Launching, firing or jettison
e. Detonation
f. Command disable

50.2 If a nuclear weapon has been selected, prearmed, or armed, and a power failure occurs, does the weapon and weapon status revert to a "deselected" condition?

50.3 Upon restoration of power, for any reason, will a previously selected, prearmed, or armed event become automatically reestablished?

50.4 Upon bringing power up, without any previous power outage, does a nuclear weapon control result in:

a. Selection
b. Prearming
c. Arming
d. Launching, firing or jettison
e. Detonation
f. Command disable

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50.5 In the event of a power failure to the nuclear weapon system command and control apparatus, does a prearmed nuclear weapon or nuclear weapon propulsion system return to the safe condition?

50.6 Will a power failure to the launcher or launching device cause the Safe/Arm device on the propulsion unit to return to SAFE?

50.7 With the restoration of power, does the Safe/Arm device ARM?

50.8 Are the performance characteristics of volatile memory elements such that power fluctuations, surges, or outages will cause erroneous or unexpected data or information to be generated? Can these variations from expected data or information cause an undesired nuclear critical function to result?

50.9 Upon restoration of power, does the weapon or weapon system automatically return to the previously selected critical function position (i.e., does a previously prearmed weapon return automatically to the prearmed condition)?

50.10 Is the power level used in the administrative device significantly less than that required for critical functions?

50.11 Can the firing system in a propulsion unit respond to a fire command without a previous arm command?

60 Launcher

60.1 Can the failure of any single component in a howitzer, gun, cannon or launcher cause a nuclear critical function to occur?

60.2 For a weapon on a launcher or in a tactical launching container, can any function such as select, prearm, arm, launch or fire occur out of sequence? (e.g., can a weapon be prearmed prior to selection command?)

60.3 Will a failure of any device in the launcher cause any nuclear weapon function to result in:

a. Selection
b. Prearm
c. Arm
d. Launch, fire or jettison

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e. Detonation
f. Command disable

60.4 Can the motion of the launcher, in its most critical mode, transmit forces, accelerations or velocities, to a nuclear weapon or nuclear weapon component in excess of the weapon or component design limits?

60.5 Can the motion induced into a nuclear weapon or propulsion system by a launcher cause:

a. Selection
b. Prearming
c. Arming
d. Launching, firing or jettison
e. Detonation
f. Command disable

60.6 Does the weapon, when in its most critical position on the launcher, interfere with any adjacent structure?

60.7 On the launcher, what is the closest point of approach of any part of the weapon (nose, control surfaces, etc.,) to any of the adjacent structures?

60.8 Upon launch, what is the closest point of approach of any part of the weapon to any of the adjacent or movable structures?

70 Masked Weapon Condition

70.1 Under "masked weapon" conditions, are interlocks activated that would prevent launching or firing of a "masked weapon"?

70.2 Is there a "masked weapon" indicator?

80. Magazines

80.1 Are assembly areas and magazines for nuclear weapons free of harmful electromagnetic radiation signals generated by equipment and machinery in close proximity?

80.2 Are assembly areas, magazines and stowage areas of sufficient size and shape to allow at least two men to be present within the area at the same time? Is the layout, arrangement and physical characteristics of the space such that each person has an unobstructed view of the other?

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80.3 Are the various locations, racks, containers, canisters or stations marked to assist in strikedown, inventory, selection or performing any individual missile operations?

80.4 Is the spray pattern of the fire protection sprinkler system within the assembly area, magazines and stowage areas arranged to preclude an umbrella effect?

80.5 What are the rapid evacuation characteristics of assembly areas and magazines?

90 Handling, Transportation and Tiedown Requirements

90.1 Are the various components of handling equipment, stowage racks, etc., designed to discriminate between conventional weapons and nuclear weapons?

90.2 Can handling equipment that is an integral part of a launcher, cannon or other type of delivery system induce velocities, accelerations or forces into a nuclear warhead or propulsion unit that would result in:

a. Degradation?

b. Activation of a critical warhead function?

c. Activation of a critical propulsion function?

90.3 With respect to nuclear demolition munitions, can the motion and forces imposed upon the munition by various non-target environments such as handling, transportation, parachuting, ground impact, hydrostatic and barometric pressure variations, etc., cause any nuclear critical function to activate?

90.4 Based upon the stockpile-to-target sequence for the nuclear weapon under review, list the various modes of transportation specified.

90.5 Except for special warfare applications, for each mode of transportation noted, is the nuclear weapon or nuclear critical component supported by the basic frame of the transportation device as opposed to being suspended or supported by slings, straps, lift arms or hydraulics?

90.6 Is the nuclear weapon or nuclear weapon component under positive restraints during transportation?

90.7 Is there a means to provide adequate static electricity grounding throughout the transportation evolution?

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90.8 What is the degree of protection from fire that is provided the nuclear weapon during transportation? Is this in keeping with the various criteria for the weapon in question?

90.9 Does the magnitude and duration of shock imposed by the transportation onto the nuclear component exceed the allowable levels as defined for these critical components?

90.10 Are there sufficient tiedown points of adequate strength provided in each element of the transportation scheme? Are the tiedown straps or chains of adequate strength?

90.11 In the transportation analysis, was the description of the road conditions (used for determination of loads, vibration and shock) the most critical in each case?

90.12 For military transportation elements (trucks, railroad, ships, etc.) are these elements compatible with their respective military specifications?

90.13 For a particular transportation element which was procured under military specification where there are areas of non-compliance, are these areas of non-compliance detrimental to nuclear safety? (e.g., braking, structural strength, trailer dynamics, etc.)

90.14 For other transportation elements which were procured under military specifications, are these transportation elements compatible with nuclear safety?

90.15 In the event that a particular handling equipment element experiences a failure, (such as electrical, hydraulic or pneumatic), does the loaded handling device maintain control of the load (up to a rated load)?

90.16 For handling equipment with hydraulic systems, are provisions made for preventing overpressure?

90.17 Is the drift rate for hydraulic systems within the specified limits?

90.18 Are the following characteristics of handling equipment within the acceptable range insofar as nuclear safety is concerned?

a. Parallel lifting systems
b. Self-centering controls
c. Positive controls
i. Synchronized operations
j. Automatic stops
k. Load and rate limits

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d. Motion limiting devices
e. Parking brakes
f. Tines and adapters
g. Creep motion
h. Fail-safe stops
l. Maximum allowable safety factor
m. Power up and down control
n. Hooks

90.19 Does the operational concept or STS document identify or suggest any commercial vehicles to be used? Are these commercial vehicles suitable from a safety standpoint and compatible with the various principles of nuclear safety?

90.20 Are the various containers used for storage and transportation in compliance with Q-STD-328 and MIL-STD-648?

90.21 Are the pallets used for nuclear weapons in compliance with their respective military specifications?

90.22 For shipboard transportation of nuclear weapons and nuclear weapon components, are the assigned stowage compartments and magazines equipped with adequate fire protection equipment? Are sprinkler heads located such that all areas of the compartment/magazine are covered?

90.23 Are all tiedown pads designed to the proper load levels?

90.24 Are there a sufficient number of tiedown pads to accommodate the design capacity of the space?

90.25 Are hatches, elevators and doorways of sufficient size to allow safe and unhindered passage of nuclear weapons, containers, pallets and other portable handling equipment?

90.26 Are hoists, booms, cranes, etc., of sufficient strength to adequately lift the nuclear weapons?

90.27 Do these hoists, booms and cranes have the proper safety devices installed?

90.28 Is the underway replenishment floatation gear for nuclear weapons of sufficient capacity to provide floatation for the nuclear weapons and handling equipment, as well as the underway replenishment gear (such as cables, block, tackle, etc.) that could be lost overboard with the nuclear weapon in the event of an underway replenishment accident?

90.29 Do the pallets or other gear that may be used in underway replenishment have sufficient shock absorbing capacity

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to prevent shock transmission to a nuclear weapon that would exceed the weapon design values?

90.30 Do air transportable delivery vehicles meet the specific nuclear safety requirements with regard to:

a. Tiedown characteristics?
b. Magnitude of forces on tiedown rings?
c. Number and location of tiedown devices?
d. Symmetry of tiedown pattern?
e. Pallet configuration?

90.31 Are all structural devices essential to air transportation and nuclear safety properly designed to the required acceleration levels and directions? Is there sufficient capacity to adequately withstand these accelerations and forces?

90.32 Is the character of the environment transmitted into the nuclear weapons (vibration, shock, temperature, etc.) such that degradation of the nuclear weapon is caused?

90.33 Do the forces and motion induced into a nuclear projectile by the various components of the cannon or loading equipment exceed the allowable design levels of the projectile?

90.34 Are provisions adequate to safely and positively down load a nuclear projectile from the breach end of a cannon?

90.35 Is the atomic demolition munition insensitive to those environmental effects caused by parachute delivery, ground impact, hydrostatic pressures and barometric pressure variations and other non-target environments?

100 Administrative Devices

100.1 Are there administrative devices internal to the weapon structure or to the tactical container?

100.2 Are these devices of such a nature that data (serial number, etc.) can be determined by applying or connecting an electronic or fiberoptic interrogation device to the weapon or tactical container?

100.3 Does this administrative device introduce any electrical, mechanical or light energy into the weapon such that any critical function could be actuated?

100.4 What nuclear safety criteria are being satisfied through the use of administrative procedures?

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110 Optics

110.1 For those elements that are sensitive to visible light, has the design been arranged so as to preclude the entry of stray unwanted light?

110.2 Can stray or undesirable light enter optic fiber links or circuits and produce a nuclear hazard, such as activation of a nuclear critical function?

120 Electrical Circuits

120.1 Are the following elements designed to minimize the influence of electromagnetic interference?

a. Wires and wiring
b. Switches
c. Cables and connectors
d. Junction points
e. Nuclear weapon monitor circuits
f. Nuclear weapon control circuits
g. Nuclear weapon release circuits
h. Nuclear weapon firing circuits

120.2 Are optical and electrical circuits properly isolated to prevent activation of a critical event due to short, improper interference, stray or residual signals?

120.3 Are wire shields used to carry current?

120.4 Is the structure used as a ground plane?

120.5 Has the criteria of MIL-STD-1512 been applied to all electroexplosive circuits?

120.6 Are nuclear consent function circuits physically and electrically isolated from all other circuits?

120.7 Are prearming control circuits and safing circuits electrically isolated from other circuits?

120.8 Are prearming power circuits physically and electrically isolated from other circuits?

120.9 Are monitor circuits so constructed and isolated that monitoring functions are independent of weapon control functions?

120.10 Are logic circuits suitably isolated and routed to prevent short circuits?

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120.11 Are critical circuits isolated from potential sources of stray electrical power?

120.12 Has there been any advantage taken of switching placed in the ground side of power circuits?

120.13 Has the following criteria been followed in the design of wiring and cables to the maximum extent possible?

a. Minimizing coupling
b. Optimizing separation
c. Wiring secured to prevent vibration and chafing
d. Proper power wiring terminals
e. Wiring for critical circuits mechanically supported

120.14 Are the electrical connectors, pin details, connector supports, etc., designed in accordance with the rules of good design practice?

120.15 Are spare pins used for mechanical support of in-line components?

120.16 Can a short circuit due to a bent pin deliver more than 100 milliamperes current to the nuclear weapon or to critical function circuits?

120.17 Are the current levels used in monitoring and testing applications limited to a functional value below the level required to actuate the most sensitive component in the nuclear weapons system?

120.18 Is the 100 milliampere limitation for nuclear weapon/warhead monitoring strictly adhered to? For continuity, are electroexplosive devices and pressure testing circuits also designed to this limitation?

120.19 Are the criteria and constraints of panel construction and packaging followed in the design?

120.20 Have the effects of electromagnetic radiation on electroexplosive devices been properly taken into account?

120.21 List the design features within the arming and fuzing systems and associated control and monitor circuits that preclude deliberate unauthorized or accidental activation of nuclear critical functions.

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130 Abnormal Environment

130.1 Evaluate the effectiveness of these design features in various abnormal environments.

130.2 In the exposure of a nuclear weapon to a specific abnormal environment, to what extent will the various safety features survive beyond the survivability of the critical arming devices or of the nuclear device?

140 Additional Safety Checklist Items

140.1 Describe the dual signal system used in arming the warhead and how these signals are independent of each other.

140.2 Have the design criteria for the arming and fuzing device been met? Are there any deviations from the required criteria? Do these deviations diminish the overall nuclear safety?

140.3 Does the automatic processing equipment design have the necessary protective elements that will prevent automatic control until all valid and correct program data have been loaded and verified?

140.4 In the event a program error has been generated due to incorrect loading, does the system have the capability to prevent the use of the data?

140.5 Is the program and system hardware designed to comply with the constraints on unauthorized changes due to hardware faults or attempts to change without proper entry procedures?

140.6 Can programs transmitted from remote sites to the weapon control sites be verified as correct prior to utilization of the transmitted data?

140.7 In the event that automata control procedures deviate from expected or proper sequencing, will the system automatically shut down?

140.8 Denote the program or software elements of the automata system that have a first level interface with nuclear weapons and those that have a second level interface.

140.9 Can diagnostic software or residual elements of such material remain in memory after completion of maintenance or diagnostic testing?

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140.10 Has the software nuclear safety analysis determined any discrepancies in the controlling software? What are the discrepancies and how do they effect nuclear safety?

140.11 What tests and training devices or elements are there for the nuclear weapon system in question?

140.12 Does any of this test and training equipment deviate from the required design criteria? Do any such deviations degrade nuclear safety?

140.13 Can any training device in this nuclear weapon system be used on a nuclear weapon?

140.14 Based upon the application of accepted human engineering concepts to the various elements of the nuclear weapon system, what are the hazardous human error features?

140.15 Can any single human error result in the activation of a nuclear critical function? Can two independent human errors result in the activation of a nuclear critical function?

140.16 Describe how the guidance signal, for guided nuclear weapons, interacts with the final arming sequence. Once the weapon is armed, will the removal of a valid guidance signal, indicating improper impact with respect to target boundaries, cause the weapon to disarm?

140.17 Are there access means in the nuclear weapon that allow for emergency render safe procedures?

140.18 Are there any criteria specified within this document or any referenced document that result in the degradation of nuclear safety? What particular requirements have a degradation effect and what element or to what degree is nuclear safety compromised?

140.19 Identify any advanced technology concepts used in the weapon system that are not provided for in these criteria. What elements of this new technology are there that can lead to nuclear safety degradation and to the violation of the four basic nuclear safety standards.

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APPENDIX B

ACRONYMS

A/D	Arm/Disarm
A&F	Arming and Fuzing
ADM	Atomic Demolition Munition
AFSC	Air Force Systems Command
AS	Air System
ASTM	American Society for Testing and Materials
BITE	Built In Test Equipment
CAL	Completely Assembled for Launch
CD	Command Disable
COD	Carrier Onboard Delivery
DOD	Department of Defense
DOE	Department of Energy
ED	Emergency Destruct
EED	Electroexplosive Device
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMP	Electromagnetic Pulse
EMR	Electromagnetic Radiation
EOD	Explosive Ordnance Disposal
E/TSD	Environment or Trajectory Sensing Device
FED STD	Federal Standard

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G	Acceleration due to gravity
GVW	Gross Vehicle Weight
HE	High Explosive
HERO	Hazards of Electromagnetic Radiation to Explosive Ordnance
ILSP	Integrated Logistics Support Plan
IOC	Initial Operational Capability
JMSNS	Justification for Major System New Starts
LLC	Limited Life Component
MCs	Military Characteristics
MIL HDBK	Military Handbook
MIL SPEC	Military Specification
MIL STD	Military Standard
NAVSEA	Naval Sea Systems Command
NBC	Nuclear Biological Chemical
NSA	National Security Agency
NSAP	National Security Agency Publications
NWEF	Naval Weapons Evaluation Facility
NWP	Nuclear Weapon Publication
NWSSG	Nuclear Weapon Safety Study Group
OIE	Optical Interference Energy
OP	Ordnance Publications
OPDD	Operational Plan Data Document
OPNAV	Office of the Chief of Naval Operations
OS	Ordnance System

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PAL	Permissive Action Link
PMs	Project Managers
PMD	Program Management Directive
RF	Radio Frequency
RMS	Root Mean Square
ROM	Read Only Memory
S/A	Safe/Arm
SAE	Society of Automotive Engineers
SNSA	Software Nuclear Safety Analysis
STS	Stockpile-to-Target Sequence
SWOP	Special Weapons Ordnance Publication
TNT	Trinitrotoluene
UNREP	Underway Replenishment
UPS	Unique Prearming Signal
USS	Unique Signal Switch
WR	War Reserve
WSESRB	Weapon System Explosive Safety Review Board

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